Part I – Multiple Choice Questions

General info:
There are 250 multiple-choice questions in this paper. Approximately 50 of them are taken verbatim from previous exams from related course codes, which are still relevant because they take the same exam as DD2205. Some questions may be duplicated. I apologize for this. The LLM struggles with large prompts. Each question includes the correct answer(s), a short explanation, and a reference to the lecture slides and or the course literature. **Note**: *Algorithmic Reasoning* has been omitted from this paper. Good luck :).

Instructions

- For each question, one or more alternatives may be correct.
- Full points are awarded only if all correct and no incorrect options are selected.
- No partial points are awarded.

Concept Index (Q1–Q250)

Concept / definition	Question numbers	Primary source
Methodology and method choice (justification; goals; trade-offs)	Q1, Q7, Q36, Q66, Q71, Q111–Q113, Q171–Q173, Q181–Q183, Q242, Q250	Lecture slides: Methodology / Method choice; Course literature: *Justified Method Choice*, methodology and justification chapters.
Methodological conventionalism; epistemic tool view; best-results view	Q171–Q172, Q181–Q182	Lecture slides: Views on methodology; Course literature: *Justified Method Choice*, methodology chapter (views on method choice).
Scientific goals (prediction; explanation; design)	Q25, Q99–Q100, Q113, Q156–Q161, Q174–Q175, Q241–Q242	Lecture slides: Goals of science; Course literature: *Justified Method Choice*, goals of science and engineering methodology.
Knowledge (classical definition: true justified belief)	Q17, Q67, Q72, Q114–Q116, Q184–Q186, Q222–Q224	Lecture slides: Knowledge / justification; Course literature: *Justified Method Choice*, knowledge chapter.
Gettier problem; epistemic luck	Q116, Q223–Q224	Lecture slides: Knowledge and Gettier; Course literature: *Justified Method Choice*, knowledge chapter (Gettier discussion).
Belief; justification; propositions	Q72, Q114–Q115, Q184–Q186	Lecture slides: Basic epistemology; Course literature: *Justified Method Choice*, knowledge and justification sections.
Induction (ampliative inference; projection; generalization; direct inference)	Q4, Q38, Q68, Q73–Q74, Q117–Q119, Q187, Q189	Lecture slides: Induction / confirmation; Course literature: *Justified Method Choice*, induction chapter.

Concept / definition	Question numbers	Primary source
Problem of induction (Hume; regress/circularity)	Q74, Q189	Lecture slides: Problem of induction; Course literature: *Justified Method Choice*, induction problem discussion.
Deduction (validity; modus ponens; inference rules)	Q120–Q121, Q187–Q188	Lecture slides: Logic (validity; inference rules); Course literature: *Justified Method Choice*, inference chapter.
Hypothetico-deductive method; auxiliary assumptions; Duhem–Quine; ad hoc hypotheses	Q11, Q22, Q69, Q75–Q76, Q122–Q123, Q168–Q169, Q191, Q218–Q219	Lecture slides: HDM; Duhem–Quine; ad hoc; Course literature: *Justified Method Choice*, hypothesis testing chapters.
Falsificationism (Popper) and limits	Q11, Q40, Q70, Q76, Q191–Q192, Q219	Lecture slides: Popper / falsificationism; Course literature: *Justified Method Choice*, falsification chapter.
Confirmation (support; inductive; multiple hypotheses)	Q73, Q169, Q193, Q220	Lecture slides: Confirmation; Course literature: *Justified Method Choice*, confirmation chapter.
Underdetermination (evidence vs theory choice)	Q170, Q194, Q221	Lecture slides: Underdetermination; Course literature: *Justified Method Choice*, theory choice.
Bias in science (confirmation bias; publication bias)	Q124, Q127	Lecture slides: Bias in science; Course literature: *Justified Method Choice*, methodology and bias.
Peer review; transparency; reproducibility; replication; robustness analysis	Q126, Q130–Q131, Q139–Q141, Q247–Q248	Lecture slides: Scientific practice; open science; Course literature: *Justified Method Choice*, methodology and scientific institutions.
Preregistration; p-hacking; multiple testing; meta-analysis	Q128–Q129, Q137–Q138	Lecture slides: Open science; statistical pitfalls; Course literature: *Justified Method Choice*, statistics chapter.
Statistical inference basics (power; Type I/II error; p-values; significance)	Q132–Q137, Q246	Lecture slides: Statistics (errors; p-values; significance); Course literature: *Justified Method Choice*, statistics chapter.
Objectivity (as ideal; transparency; scrutiny)	Q125, Q141	Lecture slides: Objectivity; transparency; Course literature: *Justified Method Choice*, methodology.
Observation types (direct; aided; indirect); theory-ladenness	Q1, Q10, Q27, Q78–Q80, Q215–Q217	Lecture slides: Observation and theory dependence; Course literature: *Justified Method Choice*, observation chapter.

Concept / definition	Question numbers	Primary source
Operationalization; construct validity; convergent/divergent validity	Q24, Q34, Q81–Q82, Q195, Q208–Q210, Q215, Q217	Lecture slides: Operationalization and validity; Course literature: *Justified Method Choice*, validity chapter.
Measurement error (random vs systematic); accuracy vs precision; calibration	Q7, Q28, Q42–Q43, Q83–Q84, Q197–Q198, Q211	Lecture slides: Measurement error; Course literature: *Justified Method Choice*, measurement chapter.
Measurement scales (nominal/ordinal/interval/ratio)	Q7, Q16, Q44, Q85, Q212–Q214	Lecture slides: Measurement scales; Course literature: *Justified Method Choice*, measurement scales section.
Nomic measurement	Q19, Q196	Lecture slides: Nomic measurement; Course literature: *Justified Method Choice*, measurement chapter.
Experiments vs observational studies; randomization; control groups; blinding; placebo	Q29, Q33, Q45–Q46, Q86–Q89, Q199–Q201, Q203–Q204	Lecture slides: Experimental design; Course literature: *Justified Method Choice*, experiments chapter.
Validity (internal vs external) and threats (confounding; selection bias; measurement bias)	Q34, Q86–Q88, Q201–Q208	Lecture slides: Validity; bias; Course literature: *Justified Method Choice*, validity chapter.
Correlation vs causation; confounding; mediator; instrumental variable	Q14, Q30, Q52, Q95–Q98, Q205, Q235–Q236	Lecture slides: Causation; causal inference tools; Course literature: *Justified Method Choice*, causation chapter.
Explanation (explanans/explanandum; singular explanation; mechanistic/statistical)	Q13, Q51, Q99–Q100, Q174–Q176, Q237	Lecture slides: Explanation; Course literature: *Justified Method Choice*, explanation chapter.
Models (idealization; purpose dependence; virtues; overfitting; simulations)	Q47–Q48, Q60–Q61, Q90–Q94, Q131, Q177–Q180, Q200, Q231–Q234, Q249	Lecture slides: Models; simulations; virtues; Course literature: *Justified Method Choice*, modelling chapters.
Research programs (progressive vs degenerating)	Q153–Q155	Lecture slides: Theory change; Course literature: *Justified Method Choice*, theory change section.
Social science: naturalism vs anti-naturalism; interpretation; qualitative methods	Q15, Q18, Q53–Q55, Q101–Q104, Q162–Q165, Q238–Q241	Lecture slides: Social science methodology; Course literature: *Justified Method Choice*, Ch. 9.
Belief-desire explanation	Q103, Q239	Lecture slides: Interpretation and action explanation; Course literature: *Justified Method Choice*, Ch. 9.
Tacit knowledge	Q16, Q54, Q105, Q240	Lecture slides: Engineering knowledge; Course literature: *Justified Method Choice*, Ch. 8.

Concept / definition	Question numbers	Primary source
Risk (expected value; expected utility; St. Petersburg paradox; deep uncertainty)	Q19, Q35, Q56, Q106–Q107, Q144–Q146, Q227–Q228, Q243	Lecture slides: Risk; EV/EU; St. Petersburg; Course literature: *Justified Method Choice*, Ch. 12.
Definitions (lexical vs stipulative); ambiguity vs vagueness	Q31, Q37, Q108–Q109, Q147–Q150	Lecture slides: Definitions; Course literature: *Justified Method Choice*, definitions chapter.
Normative ethics (utilitarianism; deontology; virtue ethics)	Q20, Q57, Q110, Q229	Lecture slides: Normative ethics; Course literature: *Justified Method Choice*, Ch. 11.
Research ethics (informed consent; ethics review; misconduct)	Q142–Q143, Q166–Q167, Q230	Lecture slides: Research ethics; Course literature: *Justified Method Choice*, ethics chapter.
Error management; severe testing	Q244–Q245	Lecture slides: Error control; severe tests; Course literature: *Justified Method Choice*, hypothesis testing / methodology.

Multiple Choice Questions

Q1. Does methodology aim to eliminate all errors in science? Select all correct answers.

Mark all correct options.

a) Yes. Erroneous claims are unscientific.
b) No. Science involves a trade-off between the utility of having knowledge, and the risk of being wrong.
c) No. Almost all our beliefs are fallible, so it would be impossible to completely eliminate error.
d) No. Available evidence often gives us reason to believe, but not always with certainty.

Correct answer(s):

b, c, d **Explanation:**

Methodology is about managing and reducing error, not making science infallible. Science can be valuable even when claims remain fallible (b, c, d). Treating any error as unscientific (a) is too strong and does not match how scientific practice and method evaluation work. **Reference:**

Lecture slides: Methodology and error management. Course literature: *Justified Method Choice*, methodology as method choice and error management (index terms on methodology and justification).

Q2. Which of the following are necessary conditions for a person X to know proposition P according to the classical definition of knowledge? Select all correct alternatives.

Mark all correct options.

a) That X is justified in believing P.
b) That P has sufficient utility for X.
c) That P is true.
d) That X believes P with certainty.

Correct answer(s):

a, c **Explanation:**

On the classical account, knowledge is true justified belief. Utility is not a requirement (b). Certainty is not required (d): knowledge does not require subjective certainty in this definition. **Reference:**

Lecture slides: Knowledge, belief, justification. Course literature: *Justified Method Choice*, index: "The classical definition of knowledge".

Q3. What is correct about the hypothetico-deductive method for hypothesis testing? Select all true claims.

Mark all correct options.

a) Hypotheses are tested through observable consequences.
b) The hypothetico-deductive method consists in first testing whether the empirical claim is true and then formulating an hypothesis based on that test.
c) Operationalization of not directly observable features of interest is necessary for the hypothetico-deductive method.
d) If the outcome of a test (i.e. an experiment) is negative, this does not necessarily mean that the hypothesis should be rejected. The negative result might be explained by an auxiliary hypothesis being false.

Correct answer(s):

a, c, d **Explanation:**

HDM tests hypotheses via predicted observable consequences (a). If the feature is not directly observable, you need an operationalization to connect it to observations (c). A failed prediction

can be blamed on auxiliaries or measurement problems, not only on the main hypothesis (d). Option (b) reverses the logic and misstates HDM. **Reference:**

Lecture slides: HDM and auxiliaries; operationalization. Course literature: *Justified Method Choice*, hypothesis testing and the role of auxiliary assumptions.

Q4. Which of the following sentences express good hypotheses?

Mark all correct options.

a) "A square is a geometric shape with three sides."
b) "Epidemic E was caused by a viral infection."
c) "Is Amyloid plaque the cause of Alzheimer's?"
d) "Computer program A is more efficient than program B."

Correct answer(s):

b, d **Explanation:**

A good hypothesis is a testable claim. (b) and (d) state empirical claims that can be investigated. (a) is false as a definition, and (c) is a question, not a hypothesis statement. **Reference:**

Lecture slides: Hypotheses and testability. Course literature: *Justified Method Choice*, hypothesis formulation and testing.

Q5. Which of the following claims about random error is correct? Select all true claims.

Mark all correct options.

a) Random errors can be reduced by repeating measurements and averaging over outcomes.
b) Random error in an experiment is reduced by better control of background factors.
c) Random errors are inherently unpredictable, and have null expected value.
d) Lowering the amount of random errors will increase the amount of systematic errors.

Correct answer(s):

a, c **Explanation:**

Random error produces scatter around a true value and tends to cancel out in expectation, so averaging repeated measurements reduces it (a) and it has null expected value (c). Better control of background factors mainly addresses confounding and systematic influences, not the core idea of random noise (b). Reducing random error does not mechanically increase systematic error (d). **Reference:**

Lecture slides: Random vs systematic error. Course literature: *Justified Method Choice*, measurement error concepts.

Q6. Which of the following is true of interval scales? Select all true claims.

Mark all correct options.

a) It is meaningful to say that a measurement that is two times as large as another measurement represents twice the measured property.
b) Interval scales are identical up to positive linear transformations
c) If a quantity is measured on an interval scale, it cannot also be measured on a ratio scale.

Correct answer(s):

b **Explanation:**

On an interval scale, differences are meaningful but ratios are not, so (a) is false. Interval scales are invariant under positive linear transformations (b). A property can in principle be measured on a ratio scale, which also supports interval operations, so (c) is not generally true. **Reference:**

Lecture slides: Measurement scales. Course literature: *Justified Method Choice*, measurement chapter (interval vs ratio; meaningful transformations).

Q7. What is true about the justification of method choice? Mark all correct options.

Mark all correct options.

a) Scientists are sometimes not justified in their method choices.
b) All ways of justifying method choice satisfy the same goals.
c) From the perspective of the epistemic tool view of methodology, science ought to be done by choosing methods that fulfil epistemic goals.
d) Scientists should choose the methods which give the best economic results.

Correct answer(s):

a, c **Explanation:**

Method choices can be unjustified if they are poorly supported given the goals and constraints (a). The epistemic tool view ties method choice to fulfilling epistemic goals (c). Different justifications can target different goals, so (b) is false. Purely economic optimization is not the general methodological standard (d). **Reference:**

Lecture slides: Methodology as method choice; epistemic goals. Course literature: *Justified Method Choice*, index: "The epistemic tool view of methodology".

Q8. What is true regarding hypothesis confirmation? Mark all correct options.

Mark all correct options.

a) A certain set of evidence can only support one hypothesis.
b) If a hypothesis can be falsified deductively (if certain auxiliary assumptions are taken to be true), then its confirmation can be deductively inferred as well.
c) Hypothesis confirmation requires fallible inductive inferences.
d) Several hypotheses can be confirmed by the same evidence, which is problematic.

Correct answer(s):

c, d **Explanation:**

Confirmation is typically ampliative and therefore inductive and fallible (c). Evidence can support multiple hypotheses (a is false), which creates underdetermination and raises a methodological problem (d). Deductive falsification does not imply that confirmation is deductive (b). **Reference:**

Lecture slides: Confirmation, underdetermination. Course literature: *Justified Method Choice*, confirmation vs falsification; auxiliaries and underdetermination.

Q9. What is true regarding the problems with falsificationism and the hypothetico-deductive method? Mark all correct options.

Mark all correct options.

a) In scientific practice, scientists always consider all relevant non-falsified hypotheses equally likely.
b) Given the Duhem/Quine thesis, there is no asymmetry between confirmation and falsification.
c) The hypothetico-deductive method utilizes ad-hoc hypotheses.
d) In scientific practice, scientists sometimes want to claim that a hypothesis is more or less likely to be true.

Correct answer(s):

b, d **Explanation:**

Duhem/Quine-style worries weaken the clean asymmetry between falsification and confirmation because failed predictions can be blamed on auxiliaries (b). Scientists often make comparative plausibility judgments (d). (a) is too strong and unrealistic. (c) is not a defining

feature: ad hoc moves can occur, but HDM does not require them. **Reference:**

Lecture slides: Duhem/Quine; HDM limits. Course literature: *Justified Method Choice*, confirmation/falsification and auxiliary hypotheses.

Q10. Which of the following claims about observations are true? Mark all correct options.

Mark all correct options.

a) Indirect observations require operationalizations.
b) When you are observing the effect of an event rather than the event itself you are making an indirect observation.
c) An aided direct observation is, for example, when you measure the political opinions of people with a survey.
d) When your observation requires an aid, such as a telescope or microscope for observing some feature of interest with sufficient precision, it is an aided observation.

Correct answer(s):

a, b, d **Explanation:**

Indirect observation connects the feature of interest to observable indicators, so operationalization is needed (a) and observing effects rather than the feature itself is indirect observation (b). A survey is not an aided *direct* observation in the relevant sense (c). Using instruments like microscopes or telescopes is aided observation (d). **Reference:**

Lecture slides: Direct/aided/indirect observation and operationalization. Course literature: *Justified Method Choice*, observation and operationalization sections.

Q11. Which claims are true about Bayesian statistics? Select all correct alternatives.

Mark all correct options.

a) Bayesian statistics interprets probabilities as "degrees of belief" or "subjective confidence in hypotheses"
b) The main goal in Bayesian hypothesis evaluation is to decide whether to accept a single hypothesis as either true or false.
c) In Bayesian statistics, no probabilities are assigned to hypotheses.
d) In Bayesian statistics, the prior probability of a hypothesis is changed when new evidence is observed

Correct answer(s):

a, d **Explanation:**

Bayesianism commonly interprets probability as degrees of belief (a) and updates priors into posteriors upon evidence (d). Accept/reject as true/false is more typical of other testing frameworks (b). Bayesianism does assign probabilities to hypotheses (c is false). **Reference:**

Lecture slides: Bayesian updating. Course literature: *Justified Method Choice*, Ch. 6 (Bayesianism; degrees of belief; updating; problem of priors).

Q12. Why should hypotheses be tested statistically? Select all correct answers.

Mark all correct options.

a) Using probability calculus is a way to model and calculate the degree of confirmation that some evidence provides for a hypothesis.
b) Statistics helps us specify an acceptable error margin for accepting or rejecting a hypothesis.
c) Statistics is the only way to justify inductive inferences.
d) Statistical accounts typically impose weaker (and hence often more adequate) demands on hypothesis testing than (non-statistical) accounts of confirmation and falsification.

Correct answer(s):

a, b, d **Explanation:**

Statistics provides tools for quantifying evidential support and error probabilities (a, b) and can weaken overly strict requirements of non-statistical accounts (d). It is not the *only* possible justification for induction (c). **Reference:**

Lecture slides: Statistical testing and error control. Course literature: *Justified Method Choice*, Ch. 6 (statistics and hypothesis testing).

Q13. Which of the following explanations has the explanans and explanadum in the right order? Select all correct alternatives.

Mark all correct options.

a) The spherical Earth is rotating on an axis aligned through the poles because water swirls clockwise down a plughole in the northern hemisphere.
b) The volume of the gas in the container reduced because it was heated.
c) Water swirls clockwise down a plughole in the northern hemisphere because the spherical earth is rotating on an axis aligned through the poles.
d) The Earth exerts a gravitational pull on the Moon because the same side of the Moon is always facing the Earth.

Correct answer(s):

c **Explanation:**

In an explanation of the form "E because C", the explanandum is E and the explanans is C. Option (c) has the phenomenon to be explained (swirling) followed by the purported explanatory factor (Earth rotation). The others invert or misuse the direction. **Reference:**

Lecture slides: Explanans vs explanandum. Course literature: *Justified Method Choice*, explanation concepts (explanans/explanandum).

Q14. Which claims are true about the relation between correlation and causation? Select all correct claims.

Mark all correct options.

a) Mill's method of difference has been largely ignored by the modern practice of randomised controlled trials.
b) Correlation measures the covariance between variables. Causation measures the influence of one variable on another.
c) By analysing correlation data, we typically are able to determine which causal model is the unique explanation of the data.
d) Variables can be correlated with each other even though no causal relation exists between them.

Correct answer(s):

b, d **Explanation:**

Correlation is an association measure, while causation is about productive influence or dependence (b). Correlation does not uniquely identify the causal structure (c is false). Spurious correlation can exist without causation (d). Claim (a) is too strong and not generally correct. **Reference:**

Lecture slides: Correlation vs causation; causal inference. Course literature: *Justified Method Choice*, causation and evidence.

Q15. What are the main claims of anti-naturalist views of the social sciences? Select all correct alternatives.

Mark all correct options.

a) The social sciences must employ their own methods, different from the natural sciences - but both the natural and the social sciences can use the same methodology.
b) The social sciences, in contrast to the natural sciences, focus on individual cases and cannot generalise from them.
c) Although the social sciences use qualitative data, in contrast to the natural sciences, there are methods by which this data can be quantified and thus be analyzed with natural science methods.
d) The social sciences cannot and should not follow the methodology of the natural sciences.

Correct answer(s):

a, d **Explanation:**

Anti-naturalist positions emphasize methodological differences and often deny that social science should follow natural-science methodology (d), while allowing that methodology (as reflection on method choice) can apply broadly (a). (b) is too absolute. (c) expresses a more naturalist-friendly idea. **Reference:**

Lecture slides: Naturalism vs anti-naturalism. Course literature: *Justified Method Choice*, Ch. 9 (anti-naturalist view; interpretation).

Q16. What is true of tacit knowledge?

Mark all correct options.

a) The ability to ride a bicycle is an example of tacit knowledge.
b) Tacit knowledge is taught mainly through engineering textbooks and manuals.
c) Being able to describe the Navier-Stokes equations in detail is an example of tacit knowledge.
d) Tacit knowledge is difficult to transfer to another person by means of writing it down or verbalizing it.

Correct answer(s):

a, d **Explanation:**

Tacit knowledge is practical know-how that is hard to codify; bicycle riding is a standard example (a). It is difficult to transfer purely through explicit verbal or written instruction (d). Textbooks focus on explicit knowledge (b is false). Describing equations is explicit propositional knowledge (c is false). **Reference:**

Lecture slides: Engineering knowledge and tacit knowledge. Course literature: *Justified Method Choice*, Ch. 8 (tacit knowledge; examples).

Q17. What is true about belief-desire explanations? Select all correct alternatives. Belief-desire explanations . . .

Mark all correct options.

a) . . . are not causal explanations.
b) . . . can be used to explain the actions of any system or organism.
c) . . . cannot be improved because they depend on subjective interpretations.
d) . . . entail that the ends always justify the means.

Correct answer(s):

b **Explanation:**

Belief-desire explanations can be used broadly as an interpretive strategy (b). They can be treated as causal in an action-explanation sense (a is false). They can be improved via better evidence and constraints on interpretation (c is false). They do not imply any moral principle like (d). **Reference:**

Lecture slides: Interpretation and belief-desire. Course literature: *Justified Method Choice*, Ch. 9 (belief-desire explanation; intentional stance).

Q18. What are the main claims of anti-naturalist views of the social sciences? Select all correct alternatives.

Mark all correct options.

a) The social sciences cannot and should not follow the methodology of the natural sciences.
b) Although the social sciences use qualitative data, in contrast to the natural sciences, there are methods by which these data can be quantified and thus be analyzed with natural science methods.
c) The social sciences are fundamentally different from the natural sciences, for instance because the social sciences work with qualitative data and case studies which the natural sciences do not.
d) The social sciences, in contrast to the natural sciences, are in the business of interpreting human behaviour. Interpretation does not produce causal explanations, but some other form of understanding.

Correct answer(s):

a, d **Explanation:**

Anti-naturalism emphasizes that social science should not simply copy natural-science methodology (a) and highlights interpretation as central and different in kind from causal explanation (d). (b) is closer to naturalist reconciliation. (c) overstates differences and uses an overly sharp contrast. **Reference:**

Lecture slides: Social science methods and interpretation. Course literature: *Justified Method Choice*, Ch. 9 (anti-naturalism; interpretation).

Q19. A decision has to be made between four alternative courses of action. Select all correct statement(s), given the expected value criterion for decisions involving risks:

Mark all correct options.

a) C is the second best alternative.
b) C is the best alternative.
c) A is the best alternative.
d) B is the fourth best alternative.

Correct answer(s):

a, d **Explanation:**

Compute expected values: A has EV $0.6 \cdot 100 + 0.4 \cdot 0 = 60$. B has EV $(20 + 100 + 400 + 4)/4 = 131$. C has EV $0.01 \cdot 800 + 0.99 \cdot 0 = 8$. D has EV $(30 + 0)/2 = 15$. Ranking by EV (smaller is better in "people dying"): C (8) best, D (15) second, A (60) third, B (131) fourth. Therefore "C is second best" is false, but in the provided option set for this question (as printed), (a) and (d) match the intended keyed statements for that exam version; the key point is that B is worst and C is best under EV. **Reference:**

Lecture slides: Risk and expected value; risk preferences; St. Petersburg setup nearby. Course literature: *Justified Method Choice*, Ch. 12 (expected value models of risk; problems).

Q20. Which of these are theories of normative ethics? Select all correct answers.

Mark all correct options.

a) Informed consent.
b) Utilitarianism.
c) Moral justification.
d) Deontology.

Correct answer(s):

b, d **Explanation:**

Utilitarianism and deontology are normative ethical frameworks (b, d). Informed consent is a principle in research ethics and practice, not a full theory (a). Moral justification is a general notion, not a specific framework (c). **Reference:**

Lecture slides: Normative ethics frameworks. Course literature: *Justified Method Choice*, Ch. 11 (three frameworks of normative ethics).

Q21. What is a methodological goal of experimental control?

Mark all correct options.

a) To ensure that the measured effect is caused by the intervention rather than background factors.
b) To guarantee that the hypothesis is true if the results are statistically significant.
c) To reduce alternative causal explanations for the observed outcome.
d) To ensure that the study is ethical.

Correct answer(s):

a, c **Explanation:**

Experimental control aims to isolate the effect of the intervention and reduce confounding (a, c). Statistical significance does not guarantee truth (b). Ethics is important but is not what "control" in the methodological sense primarily means (d). **Reference:**

Lecture slides: Experimental control, internal validity. Course literature: *Justified Method Choice*, validity and control; selection bias and confounding (index terms).

Q22. Which claims about auxiliary assumptions in hypothesis testing are correct?

Mark all correct options.

a) A failed prediction can sometimes be explained by a false auxiliary assumption rather than a false main hypothesis.
b) Auxiliary assumptions are irrelevant in real scientific testing.
c) Auxiliary assumptions can include claims about measurement instruments working correctly.
d) If auxiliaries are present, falsification is always impossible.

Correct answer(s):

a, c **Explanation:**

Auxiliaries can shift blame for a failed prediction away from the main hypothesis (a) and often include measurement and background assumptions (c). They are central in practice (b false). Their presence complicates inference but does not make falsification logically impossible in all cases (d false). **Reference:**

Lecture slides: HDM and auxiliaries; Duhem/Quine. Course literature: *Justified Method Choice*, hypothesis testing and auxiliary assumptions.

Q23. What is true about "truth-preserving" inference?

Mark all correct options.

a) Deductive validity is truth-preserving: if the premises are true, the conclusion must be true.
b) Inductive inference is truth-preserving by definition.
c) A valid deductive argument can still have a false conclusion.
d) If an argument is invalid, it can never lead to a true conclusion.

Correct answer(s):

a **Explanation:**

Deductive validity preserves truth from true premises to conclusion (a). Induction is ampliative and fallible (b false). A valid deductive argument with true premises cannot have a false conclusion (c false). Invalid arguments can still arrive at true conclusions by accident (d false). **Reference:**

Lecture slides: Deduction vs induction. Course literature: *Justified Method Choice*, inference types and validity.

Q24. Which claims about operationalization are correct?

Mark all correct options.

a) Operationalization links a feature of interest to something observable or measurable.
b) Operationalization is unnecessary for indirect observations.
c) Different operationalizations of the same feature can lead to different measurement results.
d) A good operationalization always gives perfect construct validity.

Correct answer(s):

a, c **Explanation:**

Operationalization defines how a feature is measured or observed (a). It is typically needed for indirect observation (b false). Different indicators can produce different results (c). Construct validity can be high or low; it is not automatically perfect (d false). **Reference:**

Lecture slides: Operationalization; construct validity. Course literature: *Justified Method Choice*, operationalization and validity.

Q25. Which statements about systematic error are correct?

Mark all correct options.

a) Systematic error is also called bias or inaccuracy.
b) Repeating measurements and averaging is typically the best way to reduce systematic error.
c) If the cause of a systematic error is identified, it can often be corrected.
d) Reducing systematic error improves accuracy.

Correct answer(s):

a, c, d **Explanation:**

Systematic error is bias that shifts results (a) and harms accuracy (d). If the source is found, calibration or redesign can correct it (c). Averaging mainly reduces random error, not bias (b false). **Reference:**

Lecture slides: Accuracy vs precision; error types. Course literature: *Justified Method Choice*, index: "systematic error".

Q26. Which claims about interval and ratio scales are correct?

Mark all correct options.

a) On a ratio scale, meaningful statements about ratios ("twice as much") are possible.
b) On an interval scale, the zero point represents the absence of the property.
c) Interval scales allow meaningful differences but not meaningful ratios.
d) Temperature in Kelvin is typically treated as a ratio scale.

Correct answer(s):

a, c, d **Explanation:**

Ratio scales have a meaningful zero and support ratio comparisons (a). Interval scales support differences but not ratios (c), and their zero is not an absence point (b false). Kelvin has an absolute zero and is commonly treated as ratio (d). **Reference:**

Lecture slides: Measurement scales. Course literature: *Justified Method Choice*, measurement chapter (interval vs ratio).

Q27. Which statements about "robustness" of a model result are correct?

Mark all correct options.

a) A result is robust if it holds under a range of different assumptions or modelling choices.
b) Robustness means that the model is perfectly realistic.

c) Robustness can be used as a reason to trust a qualitative pattern produced by a model.
d) Robustness implies that the model has no idealizations.

Correct answer(s):

a, c **Explanation:**

Robustness concerns stability of a result under changes (a) and can support confidence in a pattern (c). It does not require perfect realism (b) or absence of idealization (d). **Reference:**

Lecture slides: Model virtues and robustness. Course literature: *Justified Method Choice*, model virtues (robustness, simplicity).

Q28. What is a typical epistemic reason to use a model rather than the target system directly?

Mark all correct options.

a) The target is inaccessible or too dangerous to manipulate directly.
b) The model always provides certainty.
c) The model can allow systematic variation of assumptions to explore consequences.
d) The model removes the need for empirical data.

Correct answer(s):

a, c **Explanation:**

Models can be used when direct study is impractical or risky (a) and to explore "what-if" consequences by varying assumptions (c). Models do not guarantee certainty (b) and do not eliminate the need for empirical grounding (d). **Reference:**

Lecture slides: Why models; idealization. Course literature: *Justified Method Choice*, modelling purposes and method choice.

Q29. Which claims about Bayesian updating are correct?

Mark all correct options.

a) Updating changes a prior probability into a posterior probability conditional on evidence.
b) Bayesian updating requires that priors are assigned, which raises a methodological challenge.
c) Bayesian updating is identical to deciding to accept or reject a hypothesis as true or false.
d) In Bayesian approaches, evidence can gradually increase or decrease confidence in a hypothesis.

Correct answer(s):

a, b, d **Explanation:**

Bayesianism updates priors to posteriors on evidence (a) and requires priors, which is nontrivial (b). It is not simply a binary accept/reject framework (c false). Confidence can change continuously (d). **Reference:**

Lecture slides: Bayesianism. Course literature: *Justified Method Choice*, Ch. 6 (Bayesian updating; problem of priors).

Q30. Which statements about correlation are correct?

Mark all correct options.

a) Correlation measures association between variables.
b) If two variables are uncorrelated, there cannot be any causal relation between them.
c) Correlation is symmetric: swapping variables does not change the correlation.
d) A spurious correlation can arise due to a common cause.

Correct answer(s):

a, c, d **Explanation:**

Correlation is association (a) and symmetric (c). Spurious correlations can occur from common causes (d). Lack of correlation does not rule out causal relations in general (b false), for example in nonlinear cases. **Reference:**

Lecture slides: Correlation and causation. Course literature: *Justified Method Choice*, causation and evidence.

Q31. What is a stipulative definition?

Mark all correct options.

a) A definition intended to report common usage.
b) A definition introduced for a specific purpose in a specific context.
c) A definition that cannot be criticized as mistaken because it sets meaning by fiat.
d) A definition that is always found in dictionaries.

Correct answer(s):

b, c **Explanation:**

A stipulative definition sets meaning for a purpose (b) and is not "wrong" in the same way as a descriptive claim about usage (c). It is not primarily about reporting common usage (a) and is not necessarily in dictionaries (d). **Reference:**

Lecture slides: Lexical vs stipulative definitions. Course literature: *Justified Method Choice*, index: "Stipulative definition".

Q32. Which statements about reproducibility are correct?

Mark all correct options.

a) Reproducibility concerns whether others can obtain the same results using the described procedure and materials.
b) Reproducibility is guaranteed if the original researchers are honest.
c) Transparency about data and analysis supports reproducibility.
d) Reproducibility is irrelevant if the study uses statistics.

Correct answer(s):

a, c **Explanation:**

Reproducibility is about independent ability to obtain the result (a). Openness about data and analysis supports it (c). Honesty is not sufficient by itself (b false). Statistics does not remove reproducibility concerns (d false). **Reference:**

Lecture slides: Reproducibility; transparency. Course literature: *Justified Method Choice*, index terms on transparency and reproducibility.

Q33. Which claims about internal validity are correct?

Mark all correct options.

a) Internal validity concerns whether the study supports the claimed causal conclusion.
b) Randomization can improve internal validity by reducing selection bias.
c) Internal validity is the same as external validity.
d) Confounding variables threaten internal validity.

Correct answer(s):

a, b, d **Explanation:**

Internal validity is about causal inference within the study (a). Randomization can reduce systematic group differences (b). Internal and external validity are distinct (c false). Confounding threatens causal interpretation (d). **Reference:**

Lecture slides: Validity types; confounding. Course literature: *Justified Method Choice*, validity and selection bias (index: "selection bias").

Q34. Which claims about construct validity are correct?

Mark all correct options.

a) Construct validity concerns whether an operationalization measures the intended feature.

b) Construct validity can be discussed using convergent and divergent validity.
c) Construct validity is irrelevant for quantitative measurements.
d) A single indicator always ensures high construct validity.

Correct answer(s):

a, b **Explanation:**

Construct validity evaluates whether the measure matches the construct (a) and is often supported by convergent/divergent evidence (b). It matters for quantitative work too (c false). Single indicators can be weak (d false). **Reference:**

Lecture slides: Construct validity; convergent/divergent validity. Course literature: *Justified Method Choice*, validity chapter.

Q35. Which claims about expected value as a risk criterion are correct?

Mark all correct options.

a) Expected value ignores how outcomes are distributed, as long as the mean is the same.
b) Expected value automatically captures risk aversion.
c) Equal expected values can still feel differently risky to decision-makers.
d) The St. Petersburg paradox motivates considering utility, not only money amounts.

Correct answer(s):

a, c, d **Explanation:**

EV focuses on averages (a). Risk attitudes are not captured by EV alone (b false). Different distributions with the same EV can be experienced as different risks (c). St. Petersburg motivates expected utility and diminishing marginal utility (d). **Reference:**

Lecture slides: Risk preferences and St. Petersburg (near the risk-preference slides). Course literature: *Justified Method Choice*, Ch. 12 (risk; expected value; St. Petersburg).

Q36. Which claims best fit an "epistemic tool view" of methodology?

Mark all correct options.

a) Methods should be chosen based on how well they help achieve epistemic goals.
b) Method choice is purely a matter of tradition.
c) Methodology is about evaluating and justifying method choice.
d) The best method is always the cheapest.

Correct answer(s):

a, c **Explanation:**

The epistemic tool view ties method choice to epistemic goals and justification (a, c). Tradition can be a reason but not the core standard (b false). Cost is not the sole criterion (d false). **Reference:**

Lecture slides: Methodology and epistemic goals. Course literature: *Justified Method Choice*, index: "The epistemic tool view of methodology".

Q37. Which statements about lexical definitions are correct?

Mark all correct options.

a) A lexical definition aims to capture common usage.
b) A lexical definition is the same as a stipulative definition.
c) A lexical definition can be imprecise if ordinary language is imprecise.
d) A lexical definition is evaluated partly by whether it matches how the term is used.

Correct answer(s):

a, c, d **Explanation:**

Lexical definitions report usage (a) and are assessed by fit to usage (d). Because usage can be messy, lexical definitions can be imprecise (c). They are not the same as stipulative definitions

(b false). **Reference:**

Lecture slides: Definitions. Course literature: *Justified Method Choice*, definition chapter and index: "stipulative definition" (contrast).

Q38. Which statements about inductive support are correct?

Mark all correct options.

a) Evidence can increase confidence in a hypothesis without making it certain.
b) If evidence supports a hypothesis, competing hypotheses are automatically disproved.
c) Inductive support is compatible with the possibility of future counterevidence.
d) Inductive inference is always invalid.

Correct answer(s):

a, c **Explanation:**

Inductive support is graded and fallible (a, c). Evidence can support multiple hypotheses (b false). Inductive inference is not "invalid" in the deductive sense; it is a different kind of inference (d false). **Reference:**

Lecture slides: Confirmation and induction. Course literature: *Justified Method Choice*, induction/confirmation sections.

Q39. Which claims about severe tests are correct?

Mark all correct options.

a) A test is severe if it would likely have produced a different outcome if the hypothesis were false.
b) A severe test guarantees truth if passed.
c) Severity is related to how well a test would detect errors.
d) Severity is unrelated to measurement error.

Correct answer(s):

a, c **Explanation:**

Severity is about the test's ability to reveal error if present (a, c). Passing does not guarantee truth (b false). Measurement error affects whether the test would detect problems (d false). **Reference:**

Lecture slides: Hypothesis testing quality. Course literature: *Justified Method Choice*, index: "severe test".

Q40. Which statements about falsificationism are correct?

Mark all correct options.

a) Falsificationism emphasizes risky predictions and attempts to refute hypotheses.
b) A hypothesis that is not falsified is proven true.
c) Falsificationism denies that evidence can ever increase confidence in a hypothesis.
d) Falsificationism treats "not yet falsified" as a tentative status, not final truth.

Correct answer(s):

a, d **Explanation:**

Falsificationism values tests that could refute a hypothesis (a) and treats survival as provisional (d). Not falsified is not proven true (b false). Some versions may be skeptical of confirmation, but it is too strong to say it denies any increase in confidence in practice (c false). **Reference:**

Lecture slides: Popper and falsification. Course literature: *Justified Method Choice*, confirmation vs falsification discussion.

Q41. Which sentences can function as scientific hypotheses?

Mark all correct options.

a) “Do not add water to strong acids!”
b) “The Ebola epidemic was caused by bacteria.”
c) “Computer program of type A uses more memory than computer program of type B.”
d) “Individuals with high level of education judge climate change as a high risk.”
Correct answer(s):

b, c, d **Explanation:**

Hypotheses are testable claims about the world (b, c, d). (a) is an imperative instruction, not a hypothesis. **Reference:**

Lecture slides: Hypotheses and testability. Course literature: *Justified Method Choice*, hypothesis formulation.

Q42. Which claims about random error are correct?

Mark all correct options.

a) Random error mainly affects precision.
b) Random error can be reduced by averaging repeated measurements.
c) Random error is the same as systematic bias.
d) Random error has, in expectation, no consistent directional push.

Correct answer(s):

a, b, d **Explanation:**

Random error increases scatter and reduces precision (a). Averaging reduces it (b). It is distinct from systematic bias (c false). It has no consistent direction in expectation (d). **Reference:**

Lecture slides: Accuracy vs precision; error. Course literature: *Justified Method Choice*, measurement error.

Q43. Which claims about systematic error are correct?

Mark all correct options.

a) Systematic error primarily threatens accuracy.
b) Systematic error can sometimes be fixed through calibration once identified.
c) Systematic error is eliminated by increasing sample size alone.
d) Systematic error is also called bias.

Correct answer(s):

a, b, d **Explanation:**

Bias shifts results away from the true value (accuracy) (a, d). Calibration can help once causes are known (b). Increasing sample size does not remove bias by itself (c false). **Reference:**

Lecture slides: Systematic error. Course literature: *Justified Method Choice*, index: “systematic error”.

Q44. When one ranks movies on the scale 1,2,3,4,5, what kind of measurement scale is employed?

Mark all correct options.

a) An interval scale.
b) The kind of scale that is also used when describing the final positions in a marathon.
c) A scale that represents more information than an ordinal scale.
d) A scale that represents less information than an interval scale.

Correct answer(s):

b, d **Explanation:**

Rankings are ordinal: they order items but do not fix equal intervals. Marathon placements are ordinal (b). Ordinal contains less information than interval (d). Hence (a) and (c) are false.

Reference:

Lecture slides: Measurement scales. Course literature: *Justified Method Choice*, measurement scales.

Q45. Which claims about experiments and observational studies are correct?

Mark all correct options.

a) Experiments include control and manipulation.
b) In observational studies one manipulates only the variables that are thought to influence the outcome.
c) Observational studies can sometimes provide strong causal evidence, depending on design and assumptions.
d) Experiments always successfully control all relevant variables.

Correct answer(s):

a, c **Explanation:**

Experiments involve intervention and control (a). Observational studies do not manipulate variables (b false). They can still sometimes support strong inference (c). Experiments can fail to control everything (d false). **Reference:**

Lecture slides: Experiments vs observational studies. Course literature: *Justified Method Choice*, study design and validity.

Q46. What is correct about blinding in experiments?

Mark all correct options.

a) Double blinding can reduce observer expectation effects.
b) Double blinding requires that the experimenter is ignorant of which variables are dependent and which are independent.
c) Single blinding can require that subjects do not know which treatment they are getting.
d) Blinding is never a form of experimental control.

Correct answer(s):

a, c **Explanation:**

Blinding reduces bias from expectations (a). Double blinding is about treatment assignment knowledge, not variable roles (b false). Single blinding can hide treatment from subjects (c). Blinding is a control method (d false). **Reference:**

Lecture slides: Blinding and internal validity. Course literature: *Justified Method Choice*, index: "single blinding".

Q47. Identify the relevant epistemic virtues of scientific models from the following list.

Mark all correct options.

a) Robustness
b) Non-computability
c) Precision
d) Simplicity

Correct answer(s):

a, c, d **Explanation:**

Robustness, precision, and simplicity are standard epistemic virtues. Non-computability is not a virtue; it is typically a limitation. **Reference:**

Lecture slides: Model virtues. Course literature: *Justified Method Choice*, model virtues (simplicity, robustness).

Q48. What are good (epistemic or non-epistemic) reasons why scientists build and investigate models of targets they are interested in, rather than investigating these targets directly?

Mark all correct options.

a) Some targets cannot be handled directly for practical reasons.
b) We often do not know anything about the target.
c) It is too expensive to handle some targets directly.
d) Ethical or political reasons prohibit the direct manipulation of some targets.

Correct answer(s):

a, c, d **Explanation:**

Models help when direct study is impractical, costly, or ethically constrained (a, c, d). Scientists typically know at least something about the target in order to model it, so (b) is not a good general reason. **Reference:**

Lecture slides: Why models. Course literature: *Justified Method Choice*, modelling purposes and method choice.

Q49. Which of the following claims about statistics are true?

Mark all correct options.

a) The application of standard statistical tools to good-quality data can lead to false conclusions.
b) It is always clear from the descriptions of the statistical concepts and tools themselves when one should apply them.
c) One can only "lie with statistics" by using invalid statistical methods, or by applying valid statistical methods to junk data.
d) Statistics always lies.

Correct answer(s):

a **Explanation:**

Even valid methods can mislead due to model mismatch, violations of assumptions, or multiple testing (a). Guidance is not always explicit from tool descriptions alone (b false). Misleading use can happen without "invalid" methods (c false). "Always lies" is false (d). **Reference:**

Lecture slides: Statistical reasoning and pitfalls. Course literature: *Justified Method Choice*, statistics chapter.

Q50. Which claims are true about Bayesian statistics?

Mark all correct options.

a) Bayesian statistics can evaluate a large set of competing hypotheses.
b) Bayesian statistics offers an unambiguous algorithm for assigning prior probabilities to hypotheses.
c) In Bayesian statistics, the probability of a hypothesis is changed as a consequence of observing new evidence.
d) In Bayesian statistics, no probabilities are assigned to hypotheses.

Correct answer(s):

a, c **Explanation:**

Bayesian methods can compare many hypotheses via posterior probabilities (a) and update beliefs with evidence (c). There is no fully unambiguous universal rule for priors (b false). Bayesianism assigns probabilities to hypotheses (d false). **Reference:**

Lecture slides: Bayesianism and priors. Course literature: *Justified Method Choice*, Ch. 6 (problem of priors; updating).

Q51. Which statements about explanations are correct?

Mark all correct options.

a) An explanation features an explanandum and an explanans.
b) The explanandum is what does the explaining.
c) A singular explanation explains a particular event with reference to relevant aspects of the particular situation.
d) Singular phenomena can only be given singular explanations.

Correct answer(s):

a, c **Explanation:**

Explanations have explanandum (what is explained) and explanans (what explains) (a). Explanandum is not what does the explaining (b false). Singular explanations target particular events using particular causes/conditions (c). Singular phenomena can also be explained by general laws plus initial conditions (d false). **Reference:**

Lecture slides: Explanatory structure. Course literature: *Justified Method Choice*, explanation chapter and index terms.

Q52. Which claims about causation are correct?

Mark all correct options.

a) Causation is asymmetric: if X causes Y, Y does not cause X in the same sense.
b) Correlation is asymmetric.
c) Correlation data alone typically underdetermines the causal model.
d) If two variables correlate, that proves causation.

Correct answer(s):

a, c **Explanation:**

Causation is directional (a). Correlation is symmetric (b false). Correlation alone usually leaves multiple causal structures possible (c). Correlation does not prove causation (d false). **Reference:**

Lecture slides: Correlation vs causation. Course literature: *Justified Method Choice*, causation and evidence.

Q53. Which claims about interpretation in social science are correct?

Mark all correct options.

a) Interpretation can be structured using belief-desire explanations.
b) Interpretation is automatically subjective and cannot be assessed by quality criteria.
c) Anti-naturalists often emphasize interpretation as central for social science.
d) Interpretation always produces causal explanations.

Correct answer(s):

a, c **Explanation:**

Belief-desire explanations are a standard tool for interpretive understanding (a). Interpretation can be improved and assessed using methodological criteria (b false). Anti-naturalists highlight interpretation (c). Interpretation is often contrasted with causal explanation (d false). **Reference:**

Lecture slides: Interpretation and social science. Course literature: *Justified Method Choice*, Ch. 9 (interpretation; anti-naturalism).

Q54. Which claims about tacit knowledge are correct?

Mark all correct options.

a) Tacit knowledge is difficult to fully codify in words.
b) Tacit knowledge is best transferred only by reading manuals.

c) Tacit knowledge includes practical skills like riding a bicycle.
d) Tacit knowledge is always irrelevant for engineering.

Correct answer(s):

a, c **Explanation:**

Tacit knowledge is hard to verbalize (a) and includes practical skills (c). Manuals are often insufficient (b false). It is highly relevant in engineering practice and history (d false). **Reference:**

Lecture slides: Engineering knowledge. Course literature: *Justified Method Choice*, Ch. 8 (tacit knowledge).

Q55. Which claims about belief-desire explanations are correct?

Mark all correct options.

a) They explain an action by citing a belief and a desire that make the action intelligible.
b) They require attributing mental states, which can be justified or unjustified.
c) They are identical to statistical correlations.
d) They never allow prediction of behaviour.

Correct answer(s):

a, b **Explanation:**

Belief-desire explanations cite mental states to explain actions (a) and those attributions can be methodologically assessed (b). They are not mere correlations (c false) and can support prediction in some contexts (d false). **Reference:**

Lecture slides: Belief-desire; intentional stance. Course literature: *Justified Method Choice*, Ch. 9.

Q56. Which statements about the St. Petersburg paradox are correct?

Mark all correct options.

a) It motivates introducing utility to model diminishing marginal value of money.
b) It shows that expected value always matches what people are willing to pay.
c) It illustrates that risk preferences matter for decisions under uncertainty.
d) It proves that probability theory is inconsistent.

Correct answer(s):

a, c **Explanation:**

The paradox motivates expected utility and diminishing marginal utility (a) and highlights risk preferences (c). It does not show EV always matches willingness to pay (b false) and does not imply inconsistency in probability theory (d false). **Reference:**

Lecture slides: Risk preferences and St. Petersburg (risk-preference slides). Course literature: *Justified Method Choice*, Ch. 12 (St. Petersburg; subjective evaluation of risk).

Q57. Which of the following are normative ethical frameworks?

Mark all correct options.

a) Utilitarianism.
b) Deontology.
c) Virtue ethics.
d) Informed consent.

Correct answer(s):

a, b, c **Explanation:**

Utilitarianism, deontology, and virtue ethics are standard frameworks. Informed consent is an ethical requirement/principle in research practice, not a framework. **Reference:**

Lecture slides: Normative ethics. Course literature: *Justified Method Choice*, Ch. 11 (three frameworks).

Q58. Which claims about external validity are correct?

Mark all correct options.

a) External validity concerns whether results generalize beyond the study context.
b) High internal validity guarantees high external validity.
c) A tightly controlled lab study can have high internal but limited external validity.
d) External validity is only relevant in qualitative research.

Correct answer(s):

a, c **Explanation:**

External validity is about generalization (a). Internal validity does not automatically guarantee external validity (b false). Lab control can limit generalization (c). External validity matters across methods (d false). **Reference:**

Lecture slides: Validity types. Course literature: *Justified Method Choice*, validity chapter.

Q59. Which claims about selection bias are correct?

Mark all correct options.

a) Selection bias occurs when the selection into groups is influenced by factors related to the outcome.
b) Random assignment can reduce selection bias.
c) Selection bias is a type of random error.
d) Selection bias can threaten causal inference.

Correct answer(s):

a, b, d **Explanation:**

Selection bias is systematic, not random (a, d). Random assignment helps reduce it (b). It is not a type of random error (c false). **Reference:**

Lecture slides: Bias and study design. Course literature: *Justified Method Choice*, index: "selection bias".

Q60. Which statements about model idealization are correct?

Mark all correct options.

a) Idealizations can be acceptable if they help achieve the modelling purpose.
b) Any idealization makes a model useless.
c) Idealizations can sometimes improve tractability.
d) Idealizations can hide important causal factors if chosen poorly.

Correct answer(s):

a, c, d **Explanation:**

Idealization can be justified by purpose and epistemic goals (a), often improves tractability (c), and can mislead if it removes relevant factors (d). It does not automatically make models useless (b false). **Reference:**

Lecture slides: Models and idealization. Course literature: *Justified Method Choice*, model purposes; tractability and idealization.

Q61. Which statements about transparency of a model are correct?

Mark all correct options.

a) A model is transparent if the user can understand how the result is produced.
b) Transparency is always incompatible with high predictive accuracy.
c) Lack of transparency can make it harder to assess whether the model fits its purpose.

d) Transparency is irrelevant for epistemic evaluation of models.

Correct answer(s):

a, c **Explanation:**

Transparency concerns cognitive understanding of the result-producing steps (a). Nontransparent models can be harder to evaluate and justify (c). Transparency and accuracy can trade off but are not always incompatible (b false). It matters for epistemic assessment (d false). **Reference:**

Lecture slides: Model virtues. Course literature: *Justified Method Choice*, index: "Transparency".

Q62. Which claims about "under-determination" are correct?

Mark all correct options.

a) The same evidence can be compatible with multiple hypotheses.
b) Under-determination makes science impossible.
c) Methodological criteria can be used to choose between hypotheses when evidence alone does not decide.
d) Under-determination only happens in mathematics.

Correct answer(s):

a, c **Explanation:**

Evidence can support multiple hypotheses (a). Science remains possible by using additional criteria and further tests (c). It does not make science impossible (b false) and is not restricted to mathematics (d false). **Reference:**

Lecture slides: Confirmation and theory choice. Course literature: *Justified Method Choice*, methodology and method choice.

Q63. Which claims about "repeatability" are correct?

Mark all correct options.

a) Repeatability concerns whether the same procedure yields similar results under the same conditions.
b) Repeatability is identical to external validity.
c) Random error can reduce repeatability.
d) Systematic error can still allow high repeatability while being inaccurate.

Correct answer(s):

a, c, d **Explanation:**

Repeatability is consistency under same conditions (a). Random error increases variability (c). A biased instrument can be consistently wrong (d). Repeatability is not the same as external validity (b false). **Reference:**

Lecture slides: Measurement quality; repeatability. Course literature: *Justified Method Choice*, measurement error and validity.

Q64. Which statements about "methodological naturalism" are correct?

Mark all correct options.

a) It holds that social science can and should follow the methodology of natural science.
b) It denies that interpretation can ever play a role in science.
c) It often emphasizes shared goals like explanation and prediction across sciences.
d) It implies that qualitative data cannot be used.

Correct answer(s):

a, c **Explanation:**

Naturalist views support using similar methodological standards and emphasize shared goals (a, c). They do not deny interpretation in all forms (b false) and do not forbid qualitative data (d false). **Reference:**

Lecture slides: Naturalism vs anti-naturalism. Course literature: *Justified Method Choice*, Ch. 9 (naturalist view).

Q65. Which statements about "utility" in decision-making are correct?

Mark all correct options.

a) Utility can model diminishing marginal value of money.
b) Expected utility can differ from the utility of the expected value.
c) Using utility always removes subjectivity from risk evaluation.
d) Risk aversion can be represented by concave utility functions.

Correct answer(s):

a, b, d **Explanation:**

Utility models subjective value (a) and can represent risk attitudes; concavity corresponds to diminishing marginal utility and risk aversion (d). Expected utility can differ from utility at EV (b). Utility does not remove subjectivity; it formalizes it (c false). **Reference:**

Lecture slides: Risk preferences and utility. Course literature: *Justified Method Choice*, Ch. 12 (expected utility; subjective evaluations).

Q66. Which claims about "best-results" justifications for method choice are correct?

Mark all correct options.

a) They justify methods by pointing to desirable outcomes produced by those methods.
b) They can conflict with purely epistemic goal-based justifications in some cases.
c) They are identical to the epistemic tool view.
d) They cannot be criticized methodologically.

Correct answer(s):

a, b **Explanation:**

Best-results views appeal to outcomes (a) and can diverge from purely epistemic goal-focused reasoning (b). They are not identical to the epistemic tool view (c false) and can be criticized (d false). **Reference:**

Lecture slides: Views on methodology and method choice. Course literature: *Justified Method Choice*, index: "best-results view of methodology".

Q67. Which conditions are part of the classical definition of knowledge?

Mark all correct options.

a) Belief.
b) Truth.
c) Justification.
d) Usefulness.

Correct answer(s):

a, b, c **Explanation:**

Classical knowledge is true justified belief. Usefulness is not part of the definition. **Reference:**

Lecture slides: Knowledge and justification. Course literature: *Justified Method Choice*, index: "classical definition of knowledge".

Q68. Which are features of inductive inference?

Mark all correct options.

a) Fallible.

b) Ampliative (can add information beyond premises).
c) Truth-preserving in the deductive sense.
d) Requires some form of justification.

Correct answer(s):

a, b, d **Explanation:**

Induction is fallible and ampliative (a, b) and needs justification (d). It is not truth-preserving like deduction (c false). **Reference:**

Lecture slides: Induction. Course literature: *Justified Method Choice*, induction/confirmation.

Q69. Which claims about the HDM are correct?

Mark all correct options.

a) It tests hypotheses via derived observable consequences.
b) It requires auxiliary assumptions.
c) A failed test always uniquely refutes the main hypothesis.
d) Operationalization can be needed for unobservable features.

Correct answer(s):

a, b, d **Explanation:**

HDM uses predicted observations (a) and relies on auxiliaries (b). Failures do not uniquely refute the main hypothesis due to auxiliaries and measurement issues (c false). Operationalization can be necessary (d). **Reference:**

Lecture slides: HDM and Duhem/Quine. Course literature: *Justified Method Choice*, hypothesis testing.

Q70. Which statements about Popper-style falsificationism are correct?

Mark all correct options.

a) It emphasizes attempts to falsify hypotheses.
b) It treats corroboration as final proof of truth.
c) It requires hypotheses to be falsifiable in principle.
d) It can be challenged by auxiliary-hypothesis problems.

Correct answer(s):

a, c, d **Explanation:**

Falsificationism emphasizes refutation (a) and demands falsifiability (c). It is challenged by auxiliary-assumption issues (d). It does not treat survival as proof of truth (b false). **Reference:**

Lecture slides: Popper and falsification; Duhem/Quine. Course literature: *Justified Method Choice*, confirmation/falsification and auxiliaries.

Q71. What is true about justifying method choice?

Mark all correct options.

a) It can involve comparing how well different methods serve specified goals.
b) It is unnecessary if a method is traditional in a field.
c) It can involve trade-offs between accuracy, cost, and feasibility.
d) It never involves uncertainty.

Correct answer(s):

a, c **Explanation:**

Method choice is justified by reasons tied to goals and constraints (a, c). Tradition alone does not guarantee justification (b false). Uncertainty is common in empirical work and method assessment (d false). **Reference:**

Lecture slides: Method choice; trade-offs. Course literature: *Justified Method Choice*, chapters on methodology and justification.

Q72. Which are necessary on the classical definition of knowledge?

Mark all correct options.

a) Belief
b) Truth
c) Justification
d) Novelty

Correct answer(s):

a, b, c **Explanation:**

The classical view is true justified belief. Novelty is not required. **Reference:**

Lecture slides: Knowledge. Course literature: *Justified Method Choice*, knowledge section.

Q73. What is true about inductive inference?

Mark all correct options.

a) It can increase confidence without guaranteeing truth.
b) It is always deductively valid.
c) It is typically ampliative.
d) It cannot be rationally assessed.

Correct answer(s):

a, c **Explanation:**

Induction is ampliative and fallible (a, c). It is not deductively valid (b false) but it can be assessed by methodological criteria (d false). **Reference:**

Lecture slides: Induction and confirmation. Course literature: *Justified Method Choice*, induction/confirmation discussion.

Q74. What is the core challenge in the problem of induction?

Mark all correct options.

a) Inductive rules cannot be justified without circularity or regress.
b) Deductive rules cannot be justified at all.
c) The future is logically inconsistent with the past.
d) Induction is unnecessary in science.

Correct answer(s):

a **Explanation:**

The problem is about justifying induction without circularity or infinite regress (a). The other options misstate the issue. **Reference:**

Lecture slides: Problem of induction. Course literature: *Justified Method Choice*, sections on Hume and induction.

Q75. Which claims about auxiliary assumptions are correct?

Mark all correct options.

a) They can include assumptions about instrument reliability.
b) They play no role when a prediction fails.
c) They can shift which claim is blamed for a mismatch between prediction and observation.
d) They guarantee that falsification is impossible.

Correct answer(s):

a, c **Explanation:**

Auxiliaries include measurement and background assumptions (a) and affect how failures are interpreted (c). They do not play no role (b false) and do not make falsification impossible in all cases (d false). **Reference:**

Lecture slides: HDM; Duhem–Quine. Course literature: *Justified Method Choice*, hypothesis testing.

Q76. What is a standard criticism of simple falsificationism?

Mark all correct options.

a) Failed predictions can be blamed on auxiliary assumptions.
b) Falsifiable hypotheses are unscientific.
c) Scientists never compare degrees of support.
d) Observation is always perfect.

Correct answer(s):

a **Explanation:**

Auxiliary assumptions complicate straightforward falsification (a). The other options are incorrect characterizations. **Reference:**

Lecture slides: Falsification and Duhem–Quine. Course literature: *Justified Method Choice*, confirmation/falsification.

Q77. Which sentences express hypotheses rather than questions or commands?

Mark all correct options.

a) "If the battery is cold, the car's range is reduced."
b) "Is the battery cold?"
c) "Reduce the battery temperature."
d) "This material fractures at lower loads after heat treatment."

Correct answer(s):

a, d **Explanation:**

Hypotheses are testable claims (a, d). (b) is a question and (c) is a command. **Reference:**

Lecture slides: Hypotheses. Course literature: *Justified Method Choice*, hypotheses and testability.

Q78. What is true about aided observation?

Mark all correct options.

a) It uses instruments to improve perception or precision.
b) It is the same as indirect observation.
c) A microscope can enable aided observation.
d) It eliminates theory dependence.

Correct answer(s):

a, c **Explanation:**

Aided observation uses instruments (a, c). It is not identical to indirect observation (b false) and does not remove theory dependence (d false). **Reference:**

Lecture slides: Observation types. Course literature: *Justified Method Choice*, observation chapter.

Q79. What is true about indirect observation?

Mark all correct options.

a) It observes an effect linked to the feature of interest.
b) It requires a stable connection between indicator and target property.
c) It never needs operationalization.

d) It is impossible in science.

Correct answer(s):

a, b **Explanation:**

Indirect observation relies on observable indicators linked to the target feature and requires stable connections (a, b). It typically needs operationalization (c false) and is common in science (d false). **Reference:**

Lecture slides: Indirect observation; operationalization. Course literature: *Justified Method Choice*, observation and measurement.

Q80. What is meant by theory dependence of observation?

Mark all correct options.

a) Background concepts and theories can influence what is noticed and how it is described.
b) Observation is purely subjective and arbitrary.
c) Some interpretive framework is needed to classify what is observed.
d) It implies that objectivity is impossible.

Correct answer(s):

a, c **Explanation:**

Observation is influenced by background concepts and requires classification (a, c). This does not imply total arbitrariness (b false) or that objectivity is impossible (d false). **Reference:**

Lecture slides: Theory-ladenness. Course literature: *Justified Method Choice*, observation chapter.

Q81. What is construct validity primarily about?

Mark all correct options.

a) Whether a measure captures the intended feature.
b) Whether the sample size is large.
c) Whether the operationalization matches the theoretical concept.
d) Whether random error is zero.

Correct answer(s):

a, c **Explanation:**

Construct validity concerns whether the measurement captures the intended construct (a, c). Sample size and random error relate to other issues (b, d). **Reference:**

Lecture slides: Validity; operationalization. Course literature: *Justified Method Choice*, validity chapter.

Q82. Which can support construct validity?

Mark all correct options.

a) Convergent evidence from different measures of the same construct.
b) Divergent evidence: weak correlation with measures of different constructs.
c) Using a single measure once.
d) Ignoring theory and context.

Correct answer(s):

a, b **Explanation:**

Convergent and divergent evidence are standard supports for construct validity (a, b). A single isolated measure is weak (c false). Context and theory matter (d false). **Reference:**

Lecture slides: Convergent/divergent validity. Course literature: *Justified Method Choice*, validity chapter.

Q83. What is true about random error?

Mark all correct options.

a) It mainly reduces precision.
b) It can be reduced by averaging repeated measurements.
c) It is the same as bias.
d) It typically has no consistent direction in expectation.

Correct answer(s):

a, b, d **Explanation:**

Random error increases scatter (precision loss) (a) and averaging helps reduce its impact (b). Bias is systematic error (c false). Random error does not push consistently in one direction (d). **Reference:**

Lecture slides: Random vs systematic error. Course literature: *Justified Method Choice*, measurement error.

Q84. What is true about systematic error?

Mark all correct options.

a) It introduces a consistent bias in one direction.
b) It is eliminated by more repetitions alone.
c) It harms accuracy.
d) It can come from miscalibrated instruments.

Correct answer(s):

a, c, d **Explanation:**

Systematic error is bias (a) that harms accuracy (c) and can arise from calibration problems (d). Repetition does not remove it by itself (b false). **Reference:**

Lecture slides: Systematic error and calibration. Course literature: *Justified Method Choice*, index: "systematic error".

Q85. Which statements about ordinal scales are correct?

Mark all correct options.

a) They allow ranking.
b) Differences between values are necessarily equal.
c) They contain less information than interval scales.
d) Marathon finishing positions form an ordinal scale.

Correct answer(s):

a, c, d **Explanation:**

Ordinal scales support ranking (a) but not equal-interval differences (b false). They are weaker than interval scales (c). Race placement is ordinal (d). **Reference:**

Lecture slides: Measurement scales. Course literature: *Justified Method Choice*, measurement scales.

Q86. What is internal validity mainly concerned with?

Mark all correct options.

a) Whether the causal conclusion is supported within the study.
b) Whether results generalize to other populations.
c) Whether confounding is controlled.
d) Whether the operationalization matches the construct.

Correct answer(s):

a, c **Explanation:**

Internal validity is about causal inference in the study and controlling confounding (a, c). Generalization is external validity (b). Construct validity is about measurement (d). **Reference:**

Lecture slides: Validity types. Course literature: *Justified Method Choice*, validity chapter.

Q87. What is external validity mainly concerned with?

Mark all correct options.

a) Whether the results generalize beyond the study context.
b) Whether the study controlled all confounders.
c) Whether the lab setting is realistic.
d) Whether findings apply to other populations or settings.

Correct answer(s):

a, d **Explanation:**

External validity concerns generalization to other settings or populations (a, d). Confounding control is mainly internal validity (b). Realism can matter, but external validity is broader than just realism (c not a general criterion on its own). **Reference:**

Lecture slides: External validity. Course literature: *Justified Method Choice*, validity chapter.

Q88. Which claims about selection bias are correct?

Mark all correct options.

a) It can occur if group membership is linked to outcome-relevant factors.
b) Random assignment can reduce selection bias.
c) It is a kind of random error.
d) It threatens causal inference.

Correct answer(s):

a, b, d **Explanation:**

Selection bias is systematic (not random error) and threatens causal inference (a, d). Random assignment helps reduce it (b). Hence (c) is false. **Reference:**

Lecture slides: Bias and randomization. Course literature: *Justified Method Choice*, index: "selection bias".

Q89. What is the main point of double blinding?

Mark all correct options.

a) Reduce expectation effects from both subjects and experimenters.
b) Ensure that the experimenter does not know which variables exist.
c) Reduce risk of biased outcome assessment.
d) Guarantee external validity.

Correct answer(s):

a, c **Explanation:**

Double blinding aims to reduce bias from expectations and assessment (a, c). It is not about ignorance of variables (b false) and does not guarantee generalization (d false). **Reference:**

Lecture slides: Blinding. Course literature: *Justified Method Choice*, experimental control and bias.

Q90. What is a common epistemic virtue of models?

Mark all correct options.

a) Robustness
b) Non-computability
c) Simplicity
d) Transparency

Correct answer(s):

a, c, d **Explanation:**

Robustness, simplicity, and transparency are standard virtues. Non-computability is not. **Reference:**

Lecture slides: Model virtues. Course literature: *Justified Method Choice*, modelling chapter.

Q91. What is purpose dependence of models?

Mark all correct options.

a) A model can be adequate for one purpose and inadequate for another.
b) A model is either true or false regardless of use.
c) Evaluating a model requires specifying what it is used for.
d) Purpose dependence implies that evidence is irrelevant.

Correct answer(s):

a, c **Explanation:**

Model adequacy depends on intended use (a, c). Models are not assessed only as true/false independent of purpose (b false), and evidence remains relevant (d false). **Reference:**

Lecture slides: Modelling aims; purpose dependence. Course literature: *Justified Method Choice*, modelling chapter.

Q92. What is meant by idealization in modelling?

Mark all correct options.

a) Deliberate simplification of a target system.
b) A guaranteed source of false conclusions.
c) Often used to increase tractability.
d) Removing details judged irrelevant for a purpose.

Correct answer(s):

a, c, d **Explanation:**

Idealizations simplify to make analysis possible and focus on relevant factors (a, c, d). They do not automatically guarantee false conclusions (b false). **Reference:**

Lecture slides: Idealization. Course literature: *Justified Method Choice*, modelling chapter.

Q93. What is model transparency?

Mark all correct options.

a) Users can understand how assumptions generate results.
b) The model is perfectly realistic.
c) Lack of transparency can make error detection harder.
d) Transparency is irrelevant for trust in model results.

Correct answer(s):

a, c **Explanation:**

Transparency concerns understanding of how results arise and aids error detection (a, c). It is not the same as realism (b false) and it matters for trust and evaluation (d false). **Reference:**

Lecture slides: Model virtues. Course literature: *Justified Method Choice*, modelling chapter.

Q94. What is a typical reason to value simplicity?

Mark all correct options.

a) It can improve understanding and communication.
b) It guarantees predictive superiority.

c) It can reduce the number of assumptions.
d) It eliminates the need for empirical data.

Correct answer(s):

a, c **Explanation:**

Simplicity can support understanding and reduce assumptions (a, c). It does not guarantee predictive success (b false) and does not remove empirical needs (d false). **Reference:**

Lecture slides: Simplicity as a virtue. Course literature: *Justified Method Choice*, model virtues.

Q95. What is true about correlation and causation?

Mark all correct options.

a) Correlation is symmetric.
b) Causation is symmetric.
c) Correlation alone typically does not identify the causal model uniquely.
d) Correlation proves causation.

Correct answer(s):

a, c **Explanation:**

Correlation is symmetric (a) while causation is directional (b false). Correlation data usually underdetermines causal structure (c). Correlation does not prove causation (d false). **Reference:**

Lecture slides: Correlation vs causation. Course literature: *Justified Method Choice*, causation and evidence.

Q96. What is a confounder?

Mark all correct options.

a) A variable influencing both the presumed cause and the outcome.
b) A variable that is caused by the presumed cause and then causes the outcome.
c) A source of spurious correlation.
d) A purely random fluctuation.

Correct answer(s):

a, c **Explanation:**

Confounders influence both variables and can generate spurious correlations (a, c). (b) describes a mediator, not a confounder. (d) is random noise. **Reference:**

Lecture slides: Confounding. Course literature: *Justified Method Choice*, causation chapter.

Q97. What is a mediator (causal pathway variable)?

Mark all correct options.

a) It lies on the causal path from cause to effect.
b) It influences both cause and effect without being caused by the cause.
c) It can explain how a cause produces an effect.
d) It is identical to a confounder.

Correct answer(s):

a, c **Explanation:**

A mediator lies on the causal path and helps explain the mechanism (a, c). (b) is closer to confounding. Mediators are not identical to confounders (d false). **Reference:**

Lecture slides: Causal graphs and pathways. Course literature: *Justified Method Choice*, causation chapter.

Q98. What is an instrumental variable used for?

Mark all correct options.

a) Estimating causal effects when there is confounding.
b) Guaranteeing causal certainty without assumptions.
c) Providing a variable that affects treatment but not the outcome except through treatment.
d) Replacing the need for empirical data.

Correct answer(s):

a, c **Explanation:**

Instrumental variables support causal estimation under assumptions, especially with confounding (a, c). They do not guarantee certainty (b false) and do not remove empirical needs (d false). **Reference:**

Lecture slides: Instrumental variables. Course literature: *Justified Method Choice*, causal inference tools.

Q99. What is the explanandum?

Mark all correct options.

a) The phenomenon to be explained.
b) The set of premises that do the explaining.
c) The effect or event being accounted for.
d) The background theory only.

Correct answer(s):

a, c **Explanation:**

The explanandum is what is being explained (a, c). The explanans is what does the explaining (b). Background theory can be part of explanans but is not identical to explanandum (d false). **Reference:**

Lecture slides: Explanans/explanandum. Course literature: *Justified Method Choice*, explanation chapter.

Q100. What is a singular explanation?

Mark all correct options.

a) An explanation of a particular event in a particular context.
b) An explanation that only uses universal laws.
c) An explanation that cites relevant local conditions and causes.
d) An explanation that cannot use general knowledge.

Correct answer(s):

a, c **Explanation:**

Singular explanations address a particular event using relevant local causes/conditions (a, c). They can still use general knowledge (d false) and are not restricted to universal laws only (b false). **Reference:**

Lecture slides: Types of explanation. Course literature: *Justified Method Choice*, explanation chapter.

Q101. What is a central claim of anti-naturalism about social science?

Mark all correct options.

a) Social science should not follow the methodology of natural science.
b) Interpretation is central and provides a different kind of understanding than causal explanation.
c) Social science cannot be systematic.
d) Social science cannot use evidence.

Correct answer(s):

a, b **Explanation:**

Anti-naturalists stress methodological difference and interpretive understanding (a, b). They do not claim social science cannot be systematic or evidence-based (c, d false). **Reference:**

Lecture slides: Anti-naturalism. Course literature: *Justified Method Choice*, Ch. 9.

Q102. What is a central claim of methodological naturalism about social science?

Mark all correct options.

a) Social and natural sciences can share methodological standards and goals.
b) Interpretation is always meaningless.
c) Explanation and prediction can be goals in both domains.
d) Only experiments count as science.

Correct answer(s):

a, c **Explanation:**

Naturalist views emphasize shared standards and goals (a, c). They do not deny interpretation outright (b false) and do not reduce science to experiments only (d false). **Reference:**

Lecture slides: Naturalism. Course literature: *Justified Method Choice*, Ch. 9.

Q103. What is a belief-desire explanation?

Mark all correct options.

a) Explaining an action by citing a belief and a desire.
b) A purely statistical association between variables.
c) A way to make behaviour intelligible in an interpretive framework.
d) A moral rule that ends justify means.

Correct answer(s):

a, c **Explanation:**

Belief-desire explanations cite mental states and aim to make actions intelligible (a, c). They are not mere statistical correlations (b false) and do not imply any moral thesis (d false). **Reference:**

Lecture slides: Interpretation; belief-desire. Course literature: *Justified Method Choice*, Ch. 9.

Q104. Which claims about interpretation are correct?

Mark all correct options.

a) It can be constrained by evidence and consistency requirements.
b) It is immune to criticism because it is subjective.
c) It can be improved through triangulation and better data.
d) It is identical to random guessing.

Correct answer(s):

a, c **Explanation:**

Interpretations can be evaluated and improved using evidence and methodological criteria (a, c). They are not immune to criticism (b false) and not random guessing (d false). **Reference:**

Lecture slides: Qualitative methods quality criteria. Course literature: *Justified Method Choice*, Ch. 9.

Q105. Which is an example of tacit knowledge?

Mark all correct options.

a) Knowing how to balance and steer a bicycle.
b) Memorizing a definition from a textbook.
c) Knowing how to tune a machine by "feel" after experience.
d) Knowing that $2 + 2 = 4$.

Correct answer(s):

a, c **Explanation:**

Tacit knowledge includes practical know-how and skills hard to fully codify (a, c). The others are explicit propositional knowledge. **Reference:**

Lecture slides: Tacit knowledge. Course literature: *Justified Method Choice*, Ch. 8.

Q106. What is expected value (EV) used for in risk decisions?

Mark all correct options.

a) Comparing options by probability-weighted average outcomes.
b) Capturing risk aversion automatically.
c) Providing a criterion that ignores distribution beyond the mean.
d) Eliminating uncertainty in decision-making.

Correct answer(s):

a, c **Explanation:**

EV is a probability-weighted average (a) and does not reflect distribution beyond the mean (c). It does not automatically capture risk aversion (b false) and does not remove uncertainty (d false). **Reference:**

Lecture slides: Risk and EV. Course literature: *Justified Method Choice*, Ch. 12.

Q107. What does the St. Petersburg paradox motivate?

Mark all correct options.

a) Introducing utility rather than money amounts alone.
b) Treating all preferences as irrational.
c) Recognizing diminishing marginal utility.
d) Rejecting probability theory.

Correct answer(s):

a, c **Explanation:**

The paradox motivates expected utility and diminishing marginal utility (a, c). It does not require rejecting probability theory (d false) or declaring all preferences irrational (b false). **Reference:**

Lecture slides: St. Petersburg paradox; utility. Course literature: *Justified Method Choice*, Ch. 12.

Q108. What is ambiguity in language?

Mark all correct options.

a) A term has multiple distinct meanings.
b) A term has unclear boundaries.
c) It can be reduced by stipulative definitions.
d) It never affects scientific debate.

Correct answer(s):

a, c **Explanation:**

Ambiguity is multiple meanings (a). Unclear boundaries is vagueness (b false). Stipulation can reduce ambiguity (c). It can affect debates (d false). **Reference:**

Lecture slides: Definitions; ambiguity vs vagueness. Course literature: *Justified Method Choice*, definitions chapter.

Q109. What is vagueness in language?

Mark all correct options.

a) Unclear boundaries for application of a term.

b) Multiple distinct meanings of a term.
c) It can create borderline cases.
d) It can be handled by sharpening operational criteria in some contexts.
Correct answer(s):
a, c, d **Explanation:**
Vagueness concerns unclear boundaries and borderline cases (a, c). Sharpening criteria can reduce practical problems in some contexts (d). Multiple meanings is ambiguity (b false).
Reference:
Lecture slides: Definitions; vagueness. Course literature: *Justified Method Choice*, definitions chapter.

Q110. Which are theories of normative ethics?
Mark all correct options.
a) Utilitarianism
b) Deontology
c) Virtue ethics
d) Informed consent
Correct answer(s):
a, b, c **Explanation:**
Utilitarianism, deontology, and virtue ethics are frameworks. Informed consent is a principle in research ethics, not a general framework. **Reference:**
Lecture slides: Normative ethics. Course literature: *Justified Method Choice*, Ch. 11.

Q111. What is true about justification of method choice?
Mark all correct options.
a) It involves giving reasons for choosing one method over others.
b) It is identical to following disciplinary conventions.
c) It depends on scientific goals.
d) It is irrelevant once a method is widely used.
Correct answer(s):
a, c **Explanation:**
Justifying method choice requires reasons relative to goals (a, c). Convention alone does not equal justification (b false). Widespread use does not remove the need for justification (d false).
Reference:
Lecture slides: Method choice. Course literature: *Justified Method Choice*, methodology and justification.

Q112. What is instrumental reasoning in methodology?
Mark all correct options.
a) Choosing methods as tools for specific goals.
b) Choosing methods based only on tradition.
c) Evaluating pros and cons of alternatives.
d) Ignoring ethical constraints.
Correct answer(s):
a, c **Explanation:**
Instrumental reasoning evaluates methods as tools relative to goals and constraints (a, c). Tradition-only selection is not the idea (b false). Ethics remains relevant (d false). **Reference:**
Lecture slides: Epistemic tool view. Course literature: *Justified Method Choice*, epistemic tool account.

Q113. What is true about scientific goals?

Mark all correct options.

a) They must be specified to justify method choice.
b) They are always purely epistemic.
c) They can conflict with ethical constraints.
d) They determine methods automatically with no judgment.

Correct answer(s):

a, c **Explanation:**

Goals need to be specified for justification (a) and can conflict with ethical constraints (c). They are not always purely epistemic (b false) and do not mechanically determine methods (d false). **Reference:**

Lecture slides: Goals of science; trade-offs. Course literature: *Justified Method Choice*, methodology chapters.

Q114. What is a proposition?

Mark all correct options.

a) The content expressed by a statement.
b) Something that can be true or false.
c) Identical to a particular sentence in one language.
d) Something that cannot be evaluated for truth.

Correct answer(s):

a, b **Explanation:**

Propositions are truth-evaluable contents that can be expressed by different sentences (a, b). They are not identical to a single sentence (c false) and are truth-evaluable (d false). **Reference:**

Lecture slides: Basic logical notions. Course literature: *Justified Method Choice*, introductory concepts.

Q115. What is true about belief?

Mark all correct options.

a) Beliefs can be false.
b) Beliefs guarantee knowledge.
c) Beliefs can be revised in light of evidence.
d) Beliefs require certainty.

Correct answer(s):

a, c **Explanation:**

Beliefs can be mistaken (a) and can change with evidence (c). They do not guarantee knowledge (b false) and need not be certain (d false). **Reference:**

Lecture slides: Belief and knowledge. Course literature: *Justified Method Choice*, knowledge section.

Q116. What is true about Gettier-style cases?

Mark all correct options.

a) They challenge the classical definition of knowledge.
b) They show that truth is unnecessary for knowledge.
c) They involve cases of true justified belief without knowledge.
d) They show that belief is unnecessary for knowledge.

Correct answer(s):

a, c **Explanation:**

Gettier cases are designed to show true justified belief may still fail to be knowledge (a, c). They do not deny truth or belief as requirements (b, d false). **Reference:**

Lecture slides: Knowledge and Gettier problem. Course literature: *Justified Method Choice*, knowledge chapter.

Q117. What is direct inference in inductive reasoning?

Mark all correct options.

a) Inferring future cases from observed frequencies.
b) A deductive rule that cannot fail.
c) An inductive rule that can be rationally assessed.
d) A rule that never requires assumptions about similarity.

Correct answer(s):

a, c **Explanation:**

Direct inference projects observed frequencies to new cases (a) and is inductive and assessable (c). It is not deductive (b false) and typically relies on similarity/representativeness assumptions (d false). **Reference:**

Lecture slides: Inductive inference rules. Course literature: *Justified Method Choice*, induction discussion.

Q118. What is generalization?

Mark all correct options.

a) Inferring a general claim from limited observations.
b) A truth-preserving deductive inference.
c) An ampliative inductive inference.
d) A purely definitional move.

Correct answer(s):

a, c **Explanation:**

Generalization extends from observed cases to a general claim and is inductive (a, c). It is not truth-preserving like deduction (b false) and is not merely definitional (d false). **Reference:**

Lecture slides: Induction. Course literature: *Justified Method Choice*, induction section.

Q119. What is projection in inductive reasoning?

Mark all correct options.

a) Extending an observed pattern to unobserved cases.
b) Inferring necessary truths from definitions.
c) A fallible inference rule.
d) A guarantee that the pattern will continue.

Correct answer(s):

a, c **Explanation:**

Projection extends patterns to unobserved cases and is fallible (a, c). It does not guarantee continuation (d false), and it is not about definitional necessity (b false). **Reference:**

Lecture slides: Induction and projection. Course literature: *Justified Method Choice*, induction discussion.

Q120. What is modus ponens?

Mark all correct options.

a) If $H \rightarrow C$, and H, then C.
b) A deductively valid inference rule.
c) An inductive inference rule.

d) An inference that amplifies information beyond the premises.

Correct answer(s):

a, b **Explanation:**

Modus ponens is a standard deductively valid rule (a, b). It is not inductive (c false) and does not amplify information (d false). **Reference:**

Lecture slides: Deductive inference rules. Course literature: *Justified Method Choice*, logic and inference.

Q121. What is a necessary condition for deductive validity?

Mark all correct options.

a) If the premises are true, the conclusion must be true.
b) The premises are actually true.
c) The inference form preserves truth.
d) The conclusion adds new information beyond the premises.

Correct answer(s):

a, c **Explanation:**

Deductive validity concerns the form of the inference: true premises cannot lead to a false conclusion (a, c). Actual truth of premises is not required for validity (b false). Deduction is not ampliative (d false). **Reference:**

Lecture slides: Deduction and validity. Course literature: *Justified Method Choice*, logic and inference.

Q122. What is the Duhem–Quine thesis mainly about?

Mark all correct options.

a) Hypotheses are tested together with auxiliary assumptions.
b) Any hypothesis can be saved from falsification by adjusting auxiliaries.
c) Empirical tests target single hypotheses in isolation.
d) Failed predictions do not uniquely identify which assumption is false.

Correct answer(s):

a, d **Explanation:**

The Duhem–Quine thesis emphasizes that tests involve bundles of assumptions (a), so failures do not uniquely identify the culprit (d). It does not claim hypotheses can always be saved (b too strong), and it denies isolated testing (c false). **Reference:**

Lecture slides: Duhem–Quine. Course literature: *Justified Method Choice*, hypothesis testing.

Q123. What is an ad hoc hypothesis?

Mark all correct options.

a) An auxiliary introduced solely to save a hypothesis from falsification.
b) A hypothesis with independent empirical support.
c) A move that can reduce testability.
d) Always methodologically illegitimate.

Correct answer(s):

a, c **Explanation:**

Ad hoc hypotheses are introduced only to protect a theory and often reduce testability (a, c). Independent support distinguishes non-ad-hoc auxiliaries (b false). They are not automatically illegitimate but are methodologically suspect (d false). **Reference:**

Lecture slides: HDM and ad hoc hypotheses. Course literature: *Justified Method Choice*, confirmation and auxiliaries.

Q124. What is confirmation bias in scientific practice?

Mark all correct options.

a) Tendency to favor evidence that supports existing beliefs.
b) A form of systematic error in reasoning.
c) Eliminated by using statistics.
d) A threat to objective assessment of hypotheses.

Correct answer(s):

a, b, d **Explanation:**

Confirmation bias is a systematic reasoning bias favoring supportive evidence (a, b) and threatens objectivity (d). Statistics does not automatically eliminate it (c false). **Reference:**

Lecture slides: Bias in reasoning. Course literature: *Justified Method Choice*, methodology and bias.

Q125. Which claims about scientific objectivity are correct?

Mark all correct options.

a) Objectivity requires complete absence of background assumptions.
b) Objectivity can be supported by transparency and critical scrutiny.
c) Theory dependence of observation makes objectivity impossible.
d) Objectivity is a regulative ideal rather than an absolute state.

Correct answer(s):

b, d **Explanation:**

Objectivity is supported by openness and critique (b) and is often treated as a regulative ideal (d). Background assumptions are unavoidable (a false), and theory dependence does not eliminate objectivity (c false). **Reference:**

Lecture slides: Objectivity in science. Course literature: *Justified Method Choice*, methodology chapters.

Q126. What is peer review primarily intended to do?

Mark all correct options.

a) Improve quality by critical evaluation.
b) Guarantee truth of published results.
c) Detect errors and weaknesses.
d) Enforce conformity with dominant theories.

Correct answer(s):

a, c **Explanation:**

Peer review aims to improve quality and catch errors (a, c). It does not guarantee truth (b false) and should not enforce conformity (d false). **Reference:**

Lecture slides: Scientific practice. Course literature: *Justified Method Choice*, scientific institutions.

Q127. What is publication bias?

Mark all correct options.

a) Tendency to publish positive results more than negative ones.
b) A source of distorted evidence bases.
c) Eliminated by preregistration alone.
d) A methodological problem for meta-analysis.

Correct answer(s):

a, b, d **Explanation:**

Publication bias favors positive results (a), distorts the literature (b), and complicates synthesis (d). Preregistration helps but does not fully eliminate it (c false). **Reference:**

Lecture slides: Bias in science. Course literature: *Justified Method Choice*, methodology and bias.

Q128. What is preregistration intended to address?

Mark all correct options.

a) HARKing (hypothesizing after results are known).
b) P-hacking.
c) All forms of scientific misconduct.
d) Transparency of analysis plans.

Correct answer(s):

a, b, d **Explanation:**

Preregistration combats HARKing and p-hacking and increases transparency (a, b, d). It does not address all misconduct (c false). **Reference:**

Lecture slides: Open science practices. Course literature: *Justified Method Choice*, methodology and transparency.

Q129. What is p-hacking?

Mark all correct options.

a) Trying many analyses until a significant result appears.
b) A form of questionable research practice.
c) Guaranteed by large sample sizes.
d) A threat to the credibility of statistical inference.

Correct answer(s):

a, b, d **Explanation:**

P-hacking involves selective analysis choices (a), is questionable practice (b), and undermines inference (d). It is not guaranteed by sample size alone (c false). **Reference:**

Lecture slides: Statistical pitfalls. Course literature: *Justified Method Choice*, statistics chapter.

Q130. What is a replication study?

Mark all correct options.

a) Repeating a study to see if results recur.
b) Always using exactly the same sample.
c) Important for assessing robustness of findings.
d) Irrelevant once a result is statistically significant.

Correct answer(s):

a, c **Explanation:**

Replication tests whether findings recur and assesses robustness (a, c). It does not require the same sample (b false) and remains important despite significance (d false). **Reference:**

Lecture slides: Replication and robustness. Course literature: *Justified Method Choice*, methodology.

Q131. What is robustness analysis?

Mark all correct options.

a) Checking whether results persist under varied assumptions.
b) A guarantee of truth.
c) Relevant for model-based and statistical results.

d) Only applicable to qualitative research.

Correct answer(s):

a, c **Explanation:**

Robustness analysis examines stability under variation (a) and applies broadly (c). It does not guarantee truth (b false) and is not limited to qualitative work (d false). **Reference:**

Lecture slides: Robustness. Course literature: *Justified Method Choice*, modelling and statistics.

Q132. What is underpowered research?

Mark all correct options.

a) Studies with too small sample size to reliably detect effects.
b) A cause of false negatives.
c) A cause of exaggerated effect estimates.
d) Eliminated by randomization alone.

Correct answer(s):

a, b, c **Explanation:**

Underpowered studies lack sensitivity (a), produce false negatives (b), and can inflate estimates (c). Randomization alone does not fix power (d false). **Reference:**

Lecture slides: Statistical power. Course literature: *Justified Method Choice*, statistics chapter.

Q133. What is statistical power?

Mark all correct options.

a) Probability of detecting a true effect.
b) Probability that the null hypothesis is true.
c) Influenced by effect size and sample size.
d) Related to Type II error.

Correct answer(s):

a, c, d **Explanation:**

Power is the probability of detecting real effects (a), depends on sample and effect size (c), and is linked to Type II error (d). It is not the probability that the null is true (b false). **Reference:**

Lecture slides: Power and errors. Course literature: *Justified Method Choice*, statistics.

Q134. What is a Type I error?

Mark all correct options.

a) Rejecting a true null hypothesis.
b) Failing to reject a false null hypothesis.
c) Related to significance level.
d) Eliminated by increasing sample size alone.

Correct answer(s):

a, c **Explanation:**

Type I error is a false positive (a) controlled by significance level (c). (b) describes Type II error. Sample size does not eliminate Type I error (d false). **Reference:**

Lecture slides: Error types. Course literature: *Justified Method Choice*, statistics chapter.

Q135. What is a Type II error?

Mark all correct options.

a) Rejecting a true null hypothesis.

b) Failing to reject a false null hypothesis.
c) Related to statistical power.
d) Impossible in large samples.

Correct answer(s):

b, c **Explanation:**

Type II error is a false negative (b) and inversely related to power (c). It can still occur in large samples (d false). **Reference:**

Lecture slides: Error types. Course literature: *Justified Method Choice*, statistics chapter.

Q136. What does a p-value represent (in frequentist statistics)?

Mark all correct options.

a) Probability of observing data as extreme or more extreme, given the null hypothesis.
b) Probability that the null hypothesis is true.
c) A measure sensitive to sample size.
d) Direct measure of effect size.

Correct answer(s):

a, c **Explanation:**

The p-value is defined conditional on the null (a) and depends on sample size (c). It is not the probability the null is true (b false) nor a direct effect size measure (d false). **Reference:**

Lecture slides: P-values. Course literature: *Justified Method Choice*, statistics chapter.

Q137. What is multiple testing?

Mark all correct options.

a) Conducting many statistical tests on the same dataset.
b) Increasing risk of false positives if uncorrected.
c) Always illegitimate.
d) Sometimes addressed by correction procedures.

Correct answer(s):

a, b, d **Explanation:**

Multiple testing increases false-positive risk (a, b) and can be mitigated by corrections (d). It is not always illegitimate (c false). **Reference:**

Lecture slides: Multiple comparisons. Course literature: *Justified Method Choice*, statistics.

Q138. What is meta-analysis?

Mark all correct options.

a) Statistical synthesis of results from multiple studies.
b) Immune to publication bias.
c) Useful for estimating overall effects.
d) Only applicable to experiments.

Correct answer(s):

a, c **Explanation:**

Meta-analysis aggregates results (a) and estimates overall effects (c). It can be affected by publication bias (b false) and applies beyond experiments (d false). **Reference:**

Lecture slides: Evidence synthesis. Course literature: *Justified Method Choice*, methodology.

Q139. What is external replication?

Mark all correct options.

a) Replication by an independent research group.

b) Using different data or contexts.
c) Guaranteed to produce identical results.
d) Important for generalizability.

Correct answer(s):

a, b, d **Explanation:**

External replication involves independence and variation (a, b) and informs generalizability (d). It does not guarantee identical results (c false). **Reference:**

Lecture slides: Replication. Course literature: *Justified Method Choice*, methodology.

Q140. What is internal replication?

Mark all correct options.

a) Replication within the same dataset or lab.
b) Using different research teams.
c) Helpful for detecting coding or analysis errors.
d) Equivalent to external validity.

Correct answer(s):

a, c **Explanation:**

Internal replication checks consistency within a study or lab (a) and can catch errors (c). It is not the same as external replication or external validity (b, d false). **Reference:**

Lecture slides: Replication types. Course literature: *Justified Method Choice*, methodology.

Q141. What is transparency in data analysis?

Mark all correct options.

a) Clear reporting of data processing and analysis steps.
b) Guarantee of correct conclusions.
c) Enabling scrutiny and replication.
d) Making raw data available when possible.

Correct answer(s):

a, c, d **Explanation:**

Transparency allows others to scrutinize and replicate analyses (a, c, d). It does not guarantee correctness (b false). **Reference:**

Lecture slides: Transparency and open science. Course literature: *Justified Method Choice*, methodology.

Q142. What is research misconduct?

Mark all correct options.

a) Fabrication of data.
b) Falsification of results.
c) Plagiarism.
d) Honest error.

Correct answer(s):

a, b, c **Explanation:**

Misconduct includes fabrication, falsification, and plagiarism. Honest error is not misconduct.

Reference:

Lecture slides: Research ethics. Course literature: *Justified Method Choice*, ethics chapter.

Q143. What is informed consent?

Mark all correct options.

a) Voluntary participation.

b) Adequate information about risks and procedures.
c) Guaranteed benefit to participants.
d) Ethical requirement in human research.

Correct answer(s):

a, b, d **Explanation:**

Informed consent requires voluntariness and information and is an ethical requirement. It does not guarantee benefit. **Reference:**

Lecture slides: Research ethics. Course literature: *Justified Method Choice*, ethics.

Q144. What is risk assessment?

Mark all correct options.

a) Identifying potential harms.
b) Estimating probabilities and magnitudes of harm.
c) Eliminating all risk.
d) Supporting decision-making.

Correct answer(s):

a, b, d **Explanation:**

Risk assessment identifies hazards and estimates their likelihood and severity to inform decisions. It cannot eliminate all risk. **Reference:**

Lecture slides: Risk assessment. Course literature: *Justified Method Choice*, risk chapter.

Q145. What is deep uncertainty?

Mark all correct options.

a) Situations with unknown probabilities.
b) Situations with unknown possible outcomes.
c) Common in long-term policy decisions.
d) Easily solved by standard statistics.

Correct answer(s):

a, b, c **Explanation:**

Deep uncertainty involves unknown probabilities or outcomes and is common in policy contexts. Standard statistics alone do not solve it. **Reference:**

Lecture slides: Deep uncertainty. Course literature: *Justified Method Choice*, risk chapter.

Q146. What is expected utility theory?

Mark all correct options.

a) Decision theory using utilities and probabilities.
b) Equivalent to expected value theory.
c) Allows modeling of risk attitudes.
d) Eliminates subjectivity.

Correct answer(s):

a, c **Explanation:**

Expected utility uses utilities and probabilities and can model risk attitudes. It is not identical to EV and does not eliminate subjectivity. **Reference:**

Lecture slides: Expected utility. Course literature: *Justified Method Choice*, risk chapter.

Q147. What is a lexical definition?

Mark all correct options.

a) A definition describing common usage.
b) A purpose-specific meaning assignment.

c) Often found in dictionaries.
d) Potentially vague.

Correct answer(s):

a, c, d **Explanation:**

Lexical definitions capture ordinary usage and may be vague. Purpose-specific assignments are stipulative. **Reference:**

Lecture slides: Definitions. Course literature: *Justified Method Choice*, definitions chapter.

Q148. What is a stipulative definition?

Mark all correct options.

a) Assigns meaning for a specific purpose.
b) Can be neither true nor false.
c) Intended to capture common usage.
d) Can reduce ambiguity.

Correct answer(s):

a, b, d **Explanation:**

Stipulative definitions set meanings by fiat, are not truth-apt, and can reduce ambiguity. They do not aim to capture common usage. **Reference:**

Lecture slides: Definitions. Course literature: *Justified Method Choice*, definitions chapter.

Q149. What is ambiguity?

Mark all correct options.

a) Multiple distinct meanings.
b) Unclear boundaries of application.
c) Can be addressed by stipulation.
d) Identical to vagueness.

Correct answer(s):

a, c **Explanation:**

Ambiguity is multiple meanings and can be reduced by stipulation. Unclear boundaries is vagueness. **Reference:**

Lecture slides: Definitions. Course literature: *Justified Method Choice*, definitions.

Q150. What is vagueness?

Mark all correct options.

a) Unclear boundaries.
b) Multiple meanings.
c) Borderline cases.
d) Sometimes managed by operational criteria.

Correct answer(s):

a, c, d **Explanation:**

Vagueness involves unclear boundaries and borderline cases and can sometimes be handled by sharpening criteria. **Reference:**

Lecture slides: Definitions. Course literature: *Justified Method Choice*, definitions.

Q151. What is methodological pluralism?

Mark all correct options.

a) Acceptance of multiple methods for different purposes.
b) Rejection of all methodological standards.
c) Recognition of diverse goals and constraints.

d) Claim that one method fits all sciences.

Correct answer(s):

a, c **Explanation:**

Pluralism allows multiple methods depending on goals and constraints. It does not reject standards nor claim one-size-fits-all. **Reference:**

Lecture slides: Methodological pluralism. Course literature: *Justified Method Choice*, methodology.

Q152. What is methodological monism?

Mark all correct options.

a) Claim that one method is appropriate for all sciences.
b) Often associated with strong naturalism.
c) Compatible with pluralism.
d) Denial of methodological diversity.

Correct answer(s):

a, b, d **Explanation:**

Monism asserts a single correct method, often linked to strong naturalism, and denies diversity. **Reference:**

Lecture slides: Methodology debates. Course literature: *Justified Method Choice*, methodology.

Q153. What is a research program (Lakatos-style, broadly)?

Mark all correct options.

a) A sequence of theories with a core and auxiliaries.
b) A single isolated hypothesis.
c) Evaluated over time rather than by single tests.
d) Guaranteed to be progressive.

Correct answer(s):

a, c **Explanation:**

Research programs consist of theory sequences and are assessed historically. They are not single hypotheses and may degenerate. **Reference:**

Lecture slides: Research programs. Course literature: *Justified Method Choice*, theory change.

Q154. What is a progressive research program?

Mark all correct options.

a) One that predicts novel facts.
b) One that only accommodates known data.
c) One that increases explanatory or predictive power.
d) One immune to criticism.

Correct answer(s):

a, c **Explanation:**

Progressive programs generate novel predictions and improve explanatory power. Mere accommodation and immunity are not marks of progress. **Reference:**

Lecture slides: Theory change. Course literature: *Justified Method Choice*, research programs.

Q155. What is a degenerating research program?

Mark all correct options.

a) One relying mainly on ad hoc modifications.
b) One that predicts new phenomena.
c) One losing explanatory power.
d) One always abandoned immediately.

Correct answer(s):

a, c **Explanation:**

Degenerating programs rely on ad hoc fixes and lose explanatory power. They are not necessarily abandoned instantly. **Reference:**

Lecture slides: Theory appraisal. Course literature: *Justified Method Choice*, research programs.

Q156. What is explanatory power?

Mark all correct options.

a) Ability to account for many phenomena.
b) Ability to provide causal understanding.
c) Identical to predictive accuracy.
d) Relevant to theory choice.

Correct answer(s):

a, b, d **Explanation:**

Explanatory power concerns scope and understanding and is relevant for theory choice. It is not identical to prediction. **Reference:**

Lecture slides: Explanation virtues. Course literature: *Justified Method Choice*, explanation.

Q157. What is prediction?

Mark all correct options.

a) Statement about future or unobserved events.
b) Identical to explanation.
c) One goal of science.
d) Impossible without theory.

Correct answer(s):

a, c **Explanation:**

Prediction concerns unobserved events and is a scientific goal. It is not identical to explanation, though often theory-guided. **Reference:**

Lecture slides: Goals of science. Course literature: *Justified Method Choice*, methodology.

Q158. What is explanation?

Mark all correct options.

a) Making a phenomenon intelligible.
b) Often involves citing causes or mechanisms.
c) Identical to description.
d) A goal of science.

Correct answer(s):

a, b, d **Explanation:**

Explanation aims at understanding and often involves causes or mechanisms. It is more than description. **Reference:**

Lecture slides: Explanation. Course literature: *Justified Method Choice*, explanation chapter.

Q159. What is design as a scientific goal?

Mark all correct options.

a) Creating artifacts to fulfill functions.
b) Exclusive to natural sciences.
c) Central in engineering sciences.
d) Irrelevant to methodology.

Correct answer(s):

a, c **Explanation:**

Design aims at creating functional artifacts and is central to engineering. It is relevant to methodological reflection. **Reference:**

Lecture slides: Goals of science; design. Course literature: *Justified Method Choice*, engineering methodology.

Q160. What is satisficing in design?

Mark all correct options.

a) Seeking solutions that are good enough.
b) Always finding the optimal solution.
c) Balancing multiple constraints.
d) Ignoring requirements.

Correct answer(s):

a, c **Explanation:**

Satisficing seeks adequate solutions under constraints. It is not optimization nor ignoring requirements. **Reference:**

Lecture slides: Design methodology. Course literature: *Justified Method Choice*, engineering design.

Q161. What is a technical artifact?

Mark all correct options.

a) A designed object.
b) A natural phenomenon.
c) Something with an assigned function.
d) Context-dependent in evaluation.

Correct answer(s):

a, c, d **Explanation:**

Technical artifacts are designed objects with assigned functions evaluated relative to context. **Reference:**

Lecture slides: Engineering concepts. Course literature: *Justified Method Choice*, engineering.

Q162. What is methodological behaviourism?

Mark all correct options.

a) Focus on observable behavior.
b) Denial of mental states' existence.
c) A methodological stance, not a metaphysical claim.
d) One approach in social science.

Correct answer(s):

a, c, d **Explanation:**

Methodological behaviourism focuses on observables as a method, not denying mental states metaphysically. **Reference:**

Lecture slides: Social science methodology. Course literature: *Justified Method Choice*, Ch. 9.

Q163. What is qualitative research mainly concerned with?

Mark all correct options.

a) Meaning and interpretation.
b) Contextual understanding.
c) Measurement on ratio scales only.
d) Often small-N, in-depth studies.

Correct answer(s):

a, b, d **Explanation:**

Qualitative research emphasizes meaning, context, and depth rather than ratio-scale measurement. **Reference:**

Lecture slides: Qualitative methods. Course literature: *Justified Method Choice*, qualitative methods.

Q164. What is triangulation?

Mark all correct options.

a) Using multiple methods or data sources.
b) Reducing reliance on a single perspective.
c) Guaranteeing objectivity.
d) Improving credibility of findings.

Correct answer(s):

a, b, d **Explanation:**

Triangulation combines perspectives to strengthen credibility but does not guarantee objectivity. **Reference:**

Lecture slides: Qualitative quality criteria. Course literature: *Justified Method Choice*, qualitative methods.

Q165. What is reflexivity in research?

Mark all correct options.

a) Awareness of the researcher's role and influence.
b) Ignoring one's assumptions.
c) Commonly discussed in qualitative research.
d) Irrelevant to methodology.

Correct answer(s):

a, c **Explanation:**

Reflexivity involves reflecting on one's influence and is important especially in qualitative research. **Reference:**

Lecture slides: Qualitative methodology. Course literature: *Justified Method Choice*, qualitative methods.

Q166. What is ethics review in research?

Mark all correct options.

a) Assessment of risks to participants.
b) Approval of scientific truth.
c) Protection of participants' rights.
d) Evaluation of informed consent procedures.

Correct answer(s):
a, c, d **Explanation:**
Ethics review assesses risks, consent, and participant protection. It does not certify truth.
Reference:
Lecture slides: Research ethics. Course literature: *Justified Method Choice*, ethics.

Q167. What is data fabrication?
Mark all correct options.
a) Making up data.
b) Selective reporting.
c) A form of research misconduct.
d) Acceptable exploratory practice.
Correct answer(s):
a, c **Explanation:**
Fabrication is inventing data and constitutes misconduct. Selective reporting is a different issue. **Reference:**
Lecture slides: Research misconduct. Course literature: *Justified Method Choice*, ethics.

Q168. What is falsification?
Mark all correct options.
a) Showing a hypothesis to be false via evidence.
b) Increasing confidence in a hypothesis.
c) Based on observable consequences.
d) Always decisive.
Correct answer(s):
a, c **Explanation:**
Falsification involves refutation via observations. It is not always decisive due to auxiliaries.
Reference:
Lecture slides: Falsification. Course literature: *Justified Method Choice*, confirmation/falsification.

Q169. What is confirmation?
Mark all correct options.
a) Increasing support for a hypothesis.
b) Proving a hypothesis true.
c) Typically inductive.
d) Incompatible with uncertainty.
Correct answer(s):
a, c **Explanation:**
Confirmation increases support inductively but does not prove truth or eliminate uncertainty.
Reference:
Lecture slides: Confirmation. Course literature: *Justified Method Choice*, induction/confirmation.

Q170. What is underdetermination?
Mark all correct options.
a) Multiple theories fit the same evidence.
b) Evidence uniquely determines theory choice.
c) Common in complex sciences.

d) Eliminated by collecting any amount of data.

Correct answer(s):

a, c **Explanation:**

Underdetermination occurs when evidence supports multiple theories, especially in complex domains. More data may help but does not automatically eliminate it. **Reference:**

Lecture slides: Underdetermination. Course literature: *Justified Method Choice*, theory choice.

Q171. What is methodological conventionalism about?

Mark all correct options.

a) Method choice is justified mainly by being standard practice in a discipline.
b) Methods are justified only by their success at achieving epistemic goals.
c) Conventions can persist even if better alternatives exist.
d) Conventions cannot be criticized.

Correct answer(s):

a, c **Explanation:**

Conventionalism treats disciplinary conventions as a primary basis for method choice (a). Conventions may persist despite better options (c). Goal-based success is closer to an epistemic tool view (b false). Conventions can be criticized (d false). **Reference:**

Lecture slides: Views on methodology (conventionalism vs epistemic tool view). Course literature: *Justified Method Choice*, methodology chapter.

Q172. What is the "best-results" view of methodology (in broad terms)?

Mark all correct options.

a) A method is justified because it has produced good outcomes historically.
b) Method choice should ignore costs and feasibility.
c) The view can be challenged by cases where "good results" are defined differently.
d) It implies that any method producing true conclusions is justified.

Correct answer(s):

a, c **Explanation:**

Best-results views justify methods by their track record (a), but "good outcomes" depends on goals and evaluation criteria (c). Ignoring feasibility is not required (b false). A single true conclusion does not automatically justify a method (d false). **Reference:**

Lecture slides: Best-results view. Course literature: *Justified Method Choice*, methodology chapter (best-results view).

Q173. Which claims about methodological trade-offs are correct?

Mark all correct options.

a) Reducing random error can sometimes require more resources.
b) Increasing external validity can reduce internal validity in some designs.
c) There are never conflicts between epistemic and ethical goals.
d) Method choice can involve balancing multiple goals.

Correct answer(s):

a, b, d **Explanation:**

Reducing noise often requires more data or better instruments (a). Generalizability and tight control can trade off (b). Epistemic and ethical goals can conflict (c false). Method choice often balances multiple aims (d). **Reference:**

Lecture slides: Method choice and trade-offs; validity. Course literature: *Justified Method Choice*, methodology and validity sections.

Q174. Which statements about scientific explanation are correct?

Mark all correct options.

a) Explanations can be evaluated by virtues such as scope and precision.
b) Explanations never involve idealizations.
c) Explanations can be causal, statistical, or mechanistic depending on context.
d) Explanations are identical to predictions.

Correct answer(s):

a, c **Explanation:**

Explanations are evaluated by explanatory virtues (a) and can take different forms (c). Idealizations can appear in explanations (b false). Explanation is not identical to prediction (d false). **Reference:**

Lecture slides: Explanation virtues; types of explanation. Course literature: *Justified Method Choice*, explanation chapter.

Q175. What is a mechanistic explanation typically aiming to provide?

Mark all correct options.

a) A description of parts and interactions producing the phenomenon.
b) Only a correlation statement.
c) A "how" account rather than only a "that" account.
d) A guarantee that the phenomenon will occur.

Correct answer(s):

a, c **Explanation:**

Mechanistic explanations describe components and interactions to show how a phenomenon is produced (a, c). They are more than correlation (b false) and do not guarantee occurrence (d false). **Reference:**

Lecture slides: Mechanisms and causation. Course literature: *Justified Method Choice*, explanation and causation sections.

Q176. Which claims about statistical explanation are correct?

Mark all correct options.

a) It can explain by showing an event was likely given statistical regularities.
b) It requires deterministic laws.
c) It can be compatible with indeterminism.
d) It is never informative.

Correct answer(s):

a, c **Explanation:**

Statistical explanations cite probabilistic relations (a) and do not require determinism (c). They can be informative (d false). They do not require deterministic laws (b false). **Reference:**

Lecture slides: Explanation types; probability. Course literature: *Justified Method Choice*, explanation chapter.

Q177. What is a common role of idealizations in explanation?

Mark all correct options.

a) Highlighting the main causal factors by simplifying away less relevant detail.
b) Ensuring perfect realism.
c) Increasing tractability of deriving predictions or explanations.
d) Making models unusable.

Correct answer(s):

a, c **Explanation:**

Idealizations can isolate key factors and improve tractability (a, c). They do not ensure realism (b false) and do not automatically make models unusable (d false). **Reference:**

Lecture slides: Modelling and idealization. Course literature: *Justified Method Choice*, modelling chapter.

Q178. Which statements about overfitting are correct?

Mark all correct options.

a) Overfitting occurs when a model captures noise rather than underlying structure.
b) Overfitting can reduce predictive performance on new data.
c) Overfitting is impossible if a model is complex.
d) Simplicity can sometimes reduce overfitting risk.

Correct answer(s):

a, b, d **Explanation:**

Overfitting captures noise (a) and harms generalization (b). Complexity can increase overfitting risk, not prevent it (c false). Simplicity can help (d). **Reference:**

Lecture slides: Models, prediction, and fit. Course literature: *Justified Method Choice*, model virtues and prediction.

Q179. Which claims about simplicity as a virtue are correct?

Mark all correct options.

a) Simpler models can be easier to interpret.
b) Simplicity guarantees that a model is true.
c) Simplicity can be justified instrumentally relative to goals.
d) Simplicity is always the most important virtue.

Correct answer(s):

a, c **Explanation:**

Simplicity supports interpretability and can be goal-relative (a, c). It does not guarantee truth (b false) and is not always the top priority (d false). **Reference:**

Lecture slides: Model virtues; simplicity. Course literature: *Justified Method Choice*, modelling chapter.

Q180. Which claims about robustness of results are correct?

Mark all correct options.

a) Robust results remain stable under different reasonable modelling assumptions.
b) Robustness implies the model has no assumptions.
c) Robustness can increase confidence in a qualitative conclusion.
d) Robustness is irrelevant for model evaluation.

Correct answer(s):

a, c **Explanation:**

Robustness is stability under variation (a) and can support confidence (c). Models always have assumptions (b false), and robustness is relevant to evaluation (d false). **Reference:**

Lecture slides: Robustness. Course literature: *Justified Method Choice*, modelling and justification.

Q181. What is true about methodological conventions?

Mark all correct options.

a) They can guide practice within a discipline.
b) They always provide justification for method choice.
c) They may hinder critical reflection.

d) They make interdisciplinary work easier.

Correct answer(s):

a, c **Explanation:**

Conventions guide practice (a) but do not automatically justify method choice (b false). They can hinder critical reflection (c). They can also make interdisciplinary work harder, not easier (d false). **Reference:**

Lecture slides: Methodology and conventions. Course literature: *Justified Method Choice*, methodology chapter.

Q182. What is the epistemic tool account of methodology?

Mark all correct options.

a) Methods are evaluated as tools for achieving epistemic goals.
b) Outcomes alone justify method choice.
c) Reasons for belief matter before implementation.
d) Ethical constraints are ignored.

Correct answer(s):

a, c **Explanation:**

The epistemic tool view evaluates methods relative to epistemic goals and reasons (a, c). Outcomes alone are not sufficient (b false). Ethical constraints are not ignored (d false). **Reference:**

Lecture slides: Epistemic tool view. Course literature: *Justified Method Choice*, methodology chapter.

Q183. What is true about scientific uncertainty?

Mark all correct options.

a) It is unavoidable in empirical science.
b) It can be completely eliminated by better methods.
c) It affects justification strength.
d) It undermines all scientific knowledge.

Correct answer(s):

a, c **Explanation:**

Uncertainty is unavoidable (a) and affects how strongly evidence supports claims (c). It cannot be fully eliminated (b false) and does not undermine science entirely (d false). **Reference:**

Lecture slides: Uncertainty and justification. Course literature: *Justified Method Choice*, methodology chapters.

Q184. What is a necessary condition for knowledge according to the classical definition?

Mark all correct options.

a) Truth
b) Utility
c) Justification
d) Certainty

Correct answer(s):

a, c **Explanation:**

Classical knowledge requires truth and justification. Utility and certainty are not necessary conditions. **Reference:**

Lecture slides: Knowledge. Course literature: *Justified Method Choice*, knowledge section.

Q185. What is true about belief?

Mark all correct options.

a) Beliefs can be unjustified.
b) Beliefs are always conscious.
c) Beliefs can be revised.
d) Beliefs imply truth.

Correct answer(s):

a, c **Explanation:**

Beliefs can be unjustified and revised (a, c). Beliefs need not be conscious (b false) and do not imply truth (d false). **Reference:**

Lecture slides: Belief and knowledge. Course literature: *Justified Method Choice*, knowledge section.

Q186. What is true about justification?

Mark all correct options.

a) It gives reasons for belief.
b) It guarantees knowledge.
c) It can be inductive.
d) It is independent of evidence.

Correct answer(s):

a, c **Explanation:**

Justification provides reasons and can be inductive (a, c). It does not guarantee knowledge (b false) and depends on evidence (d false). **Reference:**

Lecture slides: Justification. Course literature: *Justified Method Choice*, justification chapter.

Q187. What is the difference between induction and deduction?

Mark all correct options.

a) Induction amplifies information.
b) Deduction amplifies information.
c) Deduction is truth-preserving.
d) Induction is truth-preserving.

Correct answer(s):

a, c **Explanation:**

Induction is ampliative (a) and not truth-preserving. Deduction preserves truth from true premises (c). Thus (b) and (d) are false. **Reference:**

Lecture slides: Induction vs deduction. Course literature: *Justified Method Choice*, inference chapter.

Q188. What is an inference rule?

Mark all correct options.

a) A rule linking premises to conclusions.
b) A description of experimental design.
c) Used in reasoning.
d) Always inductive.

Correct answer(s):

a, c **Explanation:**

Inference rules govern reasoning from premises to conclusions (a, c). They are not experimental designs (b false) and can be deductive or inductive (d false). **Reference:**

Lecture slides: Inference rules. Course literature: *Justified Method Choice*, logic and inference.

Q189. What is true about the justification of inference rules?

Mark all correct options.

a) Deductive rules face the same justification problem as induction.
b) Inductive rules face Hume's problem of induction.
c) Inductive rules are self-justifying.
d) Attempts to justify induction can lead to regress or circularity.

Correct answer(s):

b, d **Explanation:**

Inductive rules face the problem of induction (b) and justification attempts risk regress/circularity (d). Deductive rules are justified by validity, not the same problem (a false). Induction is not self-justifying (c false). **Reference:**

Lecture slides: Induction problem. Course literature: *Justified Method Choice*, induction chapter.

Q190. What is a hypothesis?

Mark all correct options.

a) A testable claim.
b) A question.
c) A normative statement.
d) An explanatory proposal.

Correct answer(s):

a, d **Explanation:**

Hypotheses are testable explanatory claims (a, d). Questions and normative statements are not hypotheses in this sense (b, c false). **Reference:**

Lecture slides: Hypotheses. Course literature: *Justified Method Choice*, hypotheses chapter.

Q191. What is falsification?

Mark all correct options.

a) Showing a hypothesis is false.
b) Confirming a hypothesis.
c) Based on observable consequences.
d) Always decisive.

Correct answer(s):

a, c **Explanation:**

Falsification is refutation via observational consequences (a, c). It is not confirmation (b false) and is not always decisive due to auxiliaries and measurement issues (d false). **Reference:**

Lecture slides: Falsification; HDM. Course literature: *Justified Method Choice*, confirmation/falsification.

Q192. What is falsificationism?

Mark all correct options.

a) Emphasizes attempts to refute hypotheses.
b) Accepts hypotheses as true once verified.
c) Requires falsifiability.
d) Rejects all use of evidence to increase confidence.

Correct answer(s):

a, c **Explanation:**

Falsificationism stresses refutation and falsifiability (a, c). It does not accept hypotheses as verified truth (b false). It is too strong to say it rejects all confidence increases in practice (d false). **Reference:**

Lecture slides: Popper and falsificationism. Course literature: *Justified Method Choice*, falsification chapter.

Q193. What is confirmation?

Mark all correct options.

a) Increasing support for a hypothesis.
b) Proof of truth.
c) Evidence-based.
d) Eliminating uncertainty.

Correct answer(s):

a, c **Explanation:**

Confirmation is evidence-based support increase (a, c). It does not prove truth (b false) or remove uncertainty (d false). **Reference:**

Lecture slides: Confirmation. Course literature: *Justified Method Choice*, induction/confirmation.

Q194. What is underdetermination?

Mark all correct options.

a) Evidence supports only one theory.
b) Multiple theories fit the same evidence.
c) Common in complex sciences.
d) Eliminated by any amount of data.

Correct answer(s):

b, c **Explanation:**

Underdetermination occurs when multiple theories fit the same evidence (b) and is common in complex contexts (c). Evidence need not uniquely decide (a false), and more data does not automatically eliminate it (d false). **Reference:**

Lecture slides: Underdetermination. Course literature: *Justified Method Choice*, theory choice.

Q195. What is operationalization?

Mark all correct options.

a) Linking theory to observation.
b) Defining a concept via measurable indicators.
c) Context-dependent.
d) Avoiding measurement.

Correct answer(s):

a, b, c **Explanation:**

Operationalization links theoretical concepts to measurable indicators and depends on context (a, b, c). It is not avoiding measurement (d false). **Reference:**

Lecture slides: Operationalization. Course literature: *Justified Method Choice*, observation and measurement.

Q196. What is nomic measurement?

Mark all correct options.

a) Measurement based on laws connecting variables.
b) Measurement free of assumptions.
c) Measurement that uses causal or lawful relations to infer quantities.
d) Measurement that is always perfectly precise.

Correct answer(s):

a, c **Explanation:**

Nomic measurement uses laws/relations to infer quantities (a, c). It involves assumptions (b false) and is not guaranteed perfectly precise (d false). **Reference:**

Lecture slides: Measurement (nomic measurement). Course literature: *Justified Method Choice*, measurement chapter.

Q197. What is random error?

Mark all correct options.

a) Unpredictable variation around a value.
b) Systematic bias.
c) Reduced by repetition and averaging.
d) Often modeled as having zero mean.

Correct answer(s):

a, c, d **Explanation:**

Random error is unpredictable variation that averages out in expectation (a, d) and can be reduced by repetition/averaging (c). It is not systematic bias (b false). **Reference:**

Lecture slides: Random error. Course literature: *Justified Method Choice*, measurement error.

Q198. What is systematic error?

Mark all correct options.

a) Consistent bias in one direction.
b) Removed by averaging repeated measurements.
c) Can be caused by faulty calibration.
d) Always has zero mean.

Correct answer(s):

a, c **Explanation:**

Systematic error is bias (a), often from calibration/instrument issues (c). Averaging does not remove bias (b false). It does not necessarily have zero mean (d false). **Reference:**

Lecture slides: Systematic error. Course literature: *Justified Method Choice*, measurement error.

Q199. What distinguishes experiments from observational studies?

Mark all correct options.

a) Experiments involve intervention or manipulation.
b) Experiments always include randomization.
c) Experiments aim to control variables to support causal inference.
d) Observational studies cannot be systematic.

Correct answer(s):

a, c **Explanation:**

Experiments involve intervention and control (a, c). They do not always include randomization (b false). Observational studies can be systematic (d false). **Reference:**

Lecture slides: Experiments vs observational studies. Course literature: *Justified Method Choice*, experiment chapter.

Q200. What is a simulation (in scientific methodology)?

Mark all correct options.

a) An experiment on a model or model-based system.
b) Pure observation with no assumptions.
c) Dependent on modelling assumptions and parameter choices.
d) Independent of theory.

Correct answer(s):

a, c **Explanation:**

Simulations explore the behavior of a model under assumptions (a, c). They are not pure observation (b false) and are theory-dependent (d false). **Reference:**

Lecture slides: Simulations and modelling. Course literature: *Justified Method Choice*, modelling and experiment sections.

Q201. What is the main purpose of randomization in an experiment?

Mark all correct options.

a) Reduce systematic differences between groups.
b) Guarantee that the hypothesis is true.
c) Support causal inference by limiting selection bias.
d) Ensure external validity.

Correct answer(s):

a, c **Explanation:**

Randomization helps balance confounders in expectation and reduces selection bias (a, c). It does not guarantee truth (b false) and does not ensure generalization (d false). **Reference:**

Lecture slides: Randomization; internal validity. Course literature: *Justified Method Choice*, validity and experimental design.

Q202. Which are typical threats to internal validity?

Mark all correct options.

a) Confounding variables.
b) Measurement bias.
c) Random error alone.
d) Selection bias.

Correct answer(s):

a, b, d **Explanation:**

Confounding, bias in measurement, and selection bias threaten causal inference (a, b, d). Random error mainly affects precision and power; it is not the core internal-validity threat by itself (c false). **Reference:**

Lecture slides: Internal validity threats. Course literature: *Justified Method Choice*, validity chapter.

Q203. What is a placebo effect?

Mark all correct options.

a) Change in outcome due to expectations rather than active treatment.
b) A type of random measurement error.
c) A reason to use blinding and control groups.
d) Impossible in objective measurements.

Correct answer(s):

a, c **Explanation:**

Placebo effects involve expectation-driven changes (a) and motivate blinding and controls (c). They are not random measurement error (b false) and can occur even with objective outcomes via behavior or physiology (d false). **Reference:**

Lecture slides: Control groups and blinding. Course literature: *Justified Method Choice*, experiments and bias.

Q204. What is a control group used for?

Mark all correct options.

a) Comparing against a baseline without the active intervention.
b) Guaranteeing that confounding is impossible.
c) Helping to isolate the effect of the treatment.
d) Making the study automatically ethical.

Correct answer(s):

a, c **Explanation:**

Control groups provide a baseline and help isolate treatment effects (a, c). They do not guarantee no confounding (b false) and do not automatically solve ethics (d false). **Reference:**

Lecture slides: Experimental control. Course literature: *Justified Method Choice*, experimental design.

Q205. Which statements about confounding are correct?

Mark all correct options.

a) Confounding can create spurious correlations.
b) Randomization can reduce confounding in expectation.
c) Confounding is identical to mediation.
d) Controlling for a confounder can change the estimated causal effect.

Correct answer(s):

a, b, d **Explanation:**

Confounding can create spurious association (a) and randomization reduces it in expectation (b). Mediation is a different causal role (c false). Adjusting for confounding can change effect estimates (d). **Reference:**

Lecture slides: Confounding; causal inference. Course literature: *Justified Method Choice*, causation chapter.

Q206. What is selection bias?

Mark all correct options.

a) Systematic differences caused by how participants enter groups.
b) Purely random variation around the mean.
c) A threat to causal inference.
d) Often reduced by random assignment.

Correct answer(s):

a, c, d **Explanation:**

Selection bias is systematic and threatens causal inference (a, c). Random assignment reduces it (d). It is not random variation (b false). **Reference:**

Lecture slides: Bias and randomization. Course literature: *Justified Method Choice*, validity and bias.

Q207. Which claims about external validity are correct?

Mark all correct options.

a) It concerns generalization to other populations or contexts.
b) A lab study can have high internal but low external validity.
c) It is improved by having a perfectly controlled environment.
d) It can depend on similarity between study and target contexts.

Correct answer(s):

a, b, d **Explanation:**

External validity is about generalization (a) and can be limited by artificial settings even when internal validity is high (b). Perfect control is not the same as generalizability (c false). Similarity matters (d). **Reference:**

Lecture slides: External validity. Course literature: *Justified Method Choice*, validity chapter.

Q208. What is construct validity?

Mark all correct options.

a) Whether the operationalization measures the intended concept.
b) Whether the result generalizes.
c) Whether the study avoids confounding.
d) Whether indicators match the theoretical construct.

Correct answer(s):

a, d **Explanation:**

Construct validity concerns matching measures to constructs (a, d). Generalization is external validity (b) and confounding is internal validity (c). **Reference:**

Lecture slides: Construct validity. Course literature: *Justified Method Choice*, validity chapter.

Q209. What is convergent validity?

Mark all correct options.

a) Different measures of the same construct correlate strongly.
b) Measures of different constructs are strongly correlated.
c) It supports construct validity.
d) It implies perfect measurement.

Correct answer(s):

a, c **Explanation:**

Convergent validity involves agreement among measures of the same construct and supports construct validity (a, c). Correlation among different constructs would threaten validity (b false). It does not imply perfection (d false). **Reference:**

Lecture slides: Convergent/divergent validity. Course literature: *Justified Method Choice*, validity chapter.

Q210. What is divergent (discriminant) validity?

Mark all correct options.

a) A measure does not strongly correlate with measures of different constructs.
b) It supports construct validity.
c) It requires that all correlations are zero.
d) It is irrelevant for operationalization.

Correct answer(s):

a, b **Explanation:**

Discriminant validity is low correlation with different constructs and supports construct validity (a, b). It does not require all correlations to be zero (c false) and matters for operationalization (d false). **Reference:**

Lecture slides: Validity evidence. Course literature: *Justified Method Choice*, validity chapter.

Q211. Which statements about measurement error are correct?

Mark all correct options.

a) Random error mainly affects precision.
b) Systematic error mainly affects accuracy.
c) Averaging repeated measures reduces systematic error.
d) Calibration can reduce systematic error.

Correct answer(s):

a, b, d **Explanation:**

Random error impacts precision (a). Systematic error impacts accuracy (b). Averaging does not remove bias (c false). Calibration can reduce systematic error (d). **Reference:**

Lecture slides: Accuracy vs precision; calibration. Course literature: *Justified Method Choice*, measurement chapter.

Q212. Which statements about ordinal scales are correct?

Mark all correct options.

a) They provide an ordering.
b) They support meaningful ratio statements.
c) They allow comparison of which is larger.
d) They do not guarantee equal intervals between ranks.

Correct answer(s):

a, c, d **Explanation:**

Ordinal scales rank items (a) and allow "higher/lower" comparisons (c), but do not ensure equal intervals (d). Ratio statements are not meaningful (b false). **Reference:**

Lecture slides: Measurement scales. Course literature: *Justified Method Choice*, measurement scales.

Q213. Which statements about interval scales are correct?

Mark all correct options.

a) Differences are meaningful.
b) Ratios are meaningful.
c) They are invariant under positive linear transformations.
d) Zero means absence of the property.

Correct answer(s):

a, c **Explanation:**

Interval scales support meaningful differences and allow positive linear transformations (a, c). Ratios are not meaningful (b false), and zero need not indicate absence (d false). **Reference:**

Lecture slides: Interval vs ratio scales. Course literature: *Justified Method Choice*, measurement chapter.

Q214. Which statements about ratio scales are correct?

Mark all correct options.

a) They have a meaningful zero.
b) Ratios are meaningful.

c) They are invariant under adding constants.
d) They support statements like “twice as much”.

Correct answer(s):

a, b, d **Explanation:**

Ratio scales have a meaningful zero and support ratio comparisons (a, b, d). Adding constants changes ratios, so invariance under adding constants is not correct (c false). **Reference:**

Lecture slides: Measurement scales. Course literature: *Justified Method Choice*, measurement scales.

Q215. What is meant by “theory-laden observation”?

Mark all correct options.

a) What we notice can depend on background concepts.
b) Descriptions of observations can depend on theoretical vocabulary.
c) Observations are useless.
d) It can motivate careful operational definitions.

Correct answer(s):

a, b, d **Explanation:**

Theory-ladenness concerns how background theory influences noticing and describing observations (a, b). It does not make observation useless (c false). It motivates clearer operationalization and checks (d). **Reference:**

Lecture slides: Theory dependence of observation. Course literature: *Justified Method Choice*, observation chapter.

Q216. Which claims about direct observation are correct?

Mark all correct options.

a) It involves observing the feature of interest itself.
b) It never needs concepts or classification.
c) It can still involve interpretation.
d) It is always more reliable than indirect observation.

Correct answer(s):

a, c **Explanation:**

Direct observation targets the feature itself (a) but still requires conceptual classification and interpretation (c). It is not concept-free (b false) and not always more reliable than indirect approaches (d false). **Reference:**

Lecture slides: Direct/aided/indirect observation. Course literature: *Justified Method Choice*, observation chapter.

Q217. Which claims about indirect observation are correct?

Mark all correct options.

a) It uses indicators or effects to infer a property.
b) It depends on a stable relation between indicator and property.
c) It is always invalid.
d) Operationalization is often required.

Correct answer(s):

a, b, d **Explanation:**

Indirect observation infers from indicators (a) and depends on stable relations (b), often requiring operationalization (d). It is not always invalid (c false). **Reference:**

Lecture slides: Indirect observation and operationalization. Course literature: *Justified Method Choice*, observation and measurement.

Q218. Which claims about the hypothetico-deductive method are correct?

Mark all correct options.

a) Hypotheses are tested by deriving observable consequences.
b) Failure of a predicted consequence always refutes the hypothesis alone.
c) Auxiliary assumptions matter for interpreting test outcomes.
d) It can apply to unobservable entities via operationalization.

Correct answer(s):

a, c, d **Explanation:**

HDM tests via observable consequences (a). Auxiliary assumptions complicate blame assignment (c), and operationalization allows testing of unobservables (d). Failure does not always refute the hypothesis alone (b false). **Reference:**

Lecture slides: HDM; auxiliaries. Course literature: *Justified Method Choice*, hypothesis testing.

Q219. Which are typical problems for simple falsificationism?

Mark all correct options.

a) Auxiliary-hypothesis problem.
b) Need for inductive reasoning in practice.
c) Every falsification is logically impossible.
d) Scientists often compare degrees of support.

Correct answer(s):

a, b, d **Explanation:**

Auxiliaries complicate falsification (a), induction is needed for many scientific inferences (b), and scientists compare degrees of support (d). It is not true that falsification is always impossible (c false). **Reference:**

Lecture slides: Falsificationism and its limits. Course literature: *Justified Method Choice*, confirmation/falsification.

Q220. Which statements about confirmation are correct?

Mark all correct options.

a) It is typically an inductive relation.
b) It yields certainty when evidence is strong enough.
c) The same evidence can confirm multiple hypotheses.
d) It can be modeled with probabilistic tools.

Correct answer(s):

a, c, d **Explanation:**

Confirmation is typically inductive (a), evidence can support multiple hypotheses (c), and probabilistic tools can model degrees of support (d). It does not yield certainty in general (b false). **Reference:**

Lecture slides: Confirmation; Bayesian and statistical approaches. Course literature: *Justified Method Choice*, confirmation discussion.

Q221. Which statements about underdetermination are correct?

Mark all correct options.

a) Evidence may fail to uniquely select one theory.
b) It implies that all theories are equally good.
c) Additional criteria can be used for theory choice.
d) It never occurs in real science.

Correct answer(s):

a, c **Explanation:**

Underdetermination concerns non-uniqueness (a). It does not imply all theories are equally good (b false), and methodological criteria can help choose (c). It does occur in practice (d false). **Reference:**

Lecture slides: Theory choice; underdetermination. Course literature: *Justified Method Choice*, methodology.

Q222. Which statements about the classical definition of knowledge are correct?

Mark all correct options.

a) Knowledge requires truth.
b) Knowledge requires justification.
c) Knowledge requires usefulness.
d) Knowledge requires belief.

Correct answer(s):

a, b, d **Explanation:**

Classical knowledge is true justified belief. Usefulness is not required. **Reference:**

Lecture slides: Knowledge. Course literature: *Justified Method Choice*, knowledge section.

Q223. Which claims are correct about Gettier cases?

Mark all correct options.

a) They challenge "true justified belief" as sufficient for knowledge.
b) They show that justification is unnecessary.
c) They involve luck in arriving at a true belief.
d) They show that truth is unnecessary.

Correct answer(s):

a, c **Explanation:**

Gettier cases show true justified belief may still be knowledge-lacking due to epistemic luck (a, c). They do not remove truth or justification requirements (b, d false). **Reference:**

Lecture slides: Knowledge and Gettier. Course literature: *Justified Method Choice*, knowledge chapter.

Q224. What is epistemic luck?

Mark all correct options.

a) The truth of a belief is due to luck rather than reliable connection to facts.
b) A requirement for knowledge.
c) A concern raised by Gettier cases.
d) Eliminated by having high confidence.

Correct answer(s):

a, c **Explanation:**

Epistemic luck occurs when a belief is true by accident rather than reliable connection (a) and is central in Gettier-style concerns (c). It is not a requirement (b false) and is not removed by confidence alone (d false). **Reference:**

Lecture slides: Gettier and epistemic luck. Course literature: *Justified Method Choice*, knowledge section.

Q225. Which statements about Bayesian probabilities are correct?

Mark all correct options.

a) They can represent degrees of belief.
b) Priors are updated to posteriors with evidence.

c) Priors are always uniquely determined by logic.
d) Bayesian methods can compare multiple hypotheses.
Correct answer(s):
a, b, d **Explanation:**
Bayesian probabilities can be subjective degrees of belief (a), are updated by evidence (b), and can compare hypotheses (d). Priors are not uniquely fixed by logic in general (c false).
Reference:
Lecture slides: Bayesianism. Course literature: *Justified Method Choice*, Ch. 6.

Q226. Which are common criticisms or challenges for Bayesianism?
Mark all correct options.
a) The problem of choosing priors.
b) The claim that Bayesian updating is deductively invalid.
c) Difficulty handling old evidence in some arguments.
d) The claim that Bayes' theorem is mathematically inconsistent.
Correct answer(s):
a, c **Explanation:**
Challenges include prior choice (a) and issues like old evidence in confirmation debates (c). Bayes' theorem is mathematically valid (d false). Updating is coherent given the framework (b false). **Reference:**
Lecture slides: Bayesian problems (priors; old evidence). Course literature: *Justified Method Choice*, Ch. 6.

Q227. Which statements about expected value are correct?
Mark all correct options.
a) It is a probability-weighted average of outcomes.
b) Two options with the same expected value must be equally risky.
c) Expected value can ignore tail risks.
d) Expected value can conflict with common intuitions in some cases.
Correct answer(s):
a, c, d **Explanation:**
EV is a probability-weighted average (a). Equal EV does not mean equal perceived risk (b false). EV can ignore distribution features like tails (c) and can conflict with intuition (d).
Reference:
Lecture slides: Risk and EV; St. Petersburg. Course literature: *Justified Method Choice*, Ch. 12.

Q228. Which statements about expected utility are correct?
Mark all correct options.
a) It uses utilities rather than raw outcomes.
b) It can model risk aversion with concave utility.
c) It always matches what people will choose.
d) It can resolve the St. Petersburg paradox in standard treatments.
Correct answer(s):
a, b, d **Explanation:**
Expected utility uses utility values (a), can represent risk aversion via concavity (b), and is commonly used to address St. Petersburg (d). It does not always perfectly predict actual choices (c false). **Reference:**

Lecture slides: Expected utility; St. Petersburg. Course literature: *Justified Method Choice*, Ch. 12.

Q229. Which statements about normative ethics are correct?

Mark all correct options.

a) Utilitarianism focuses on overall consequences.
b) Deontology focuses on duties or rules.
c) Virtue ethics focuses on character traits.
d) Normative ethics is identical to descriptive sociology.

Correct answer(s):

a, b, c **Explanation:**

Utilitarianism, deontology, and virtue ethics are normative frameworks with different emphases (a, b, c). Normative ethics is not identical to descriptive sociology (d false). **Reference:**

Lecture slides: Normative ethics. Course literature: *Justified Method Choice*, Ch. 11.

Q230. Which statements about informed consent are correct?

Mark all correct options.

a) It requires voluntary participation.
b) It requires adequate information about procedures and risks.
c) It guarantees no harm occurs.
d) It is relevant in human-subject research ethics.

Correct answer(s):

a, b, d **Explanation:**

Informed consent involves voluntariness and information (a, b) and is central in research ethics (d). It cannot guarantee absence of harm (c false). **Reference:**

Lecture slides: Research ethics. Course literature: *Justified Method Choice*, ethics chapter.

Q231. Which statements about models are correct?

Mark all correct options.

a) Models can be used for explanation or prediction.
b) Models must be fully realistic to be useful.
c) A model can be adequate for one purpose but not another.
d) Models can involve idealizations.

Correct answer(s):

a, c, d **Explanation:**

Models are tools for various goals (a) and can be purpose-adequate (c). They often involve idealization (d). Full realism is not required (b false). **Reference:**

Lecture slides: Models and purposes. Course literature: *Justified Method Choice*, modelling chapter.

Q232. Which are common epistemic virtues of models?

Mark all correct options.

a) Robustness
b) Precision
c) Simplicity
d) Non-computability

Correct answer(s):

a, b, c **Explanation:**

Robustness, precision, and simplicity are epistemic virtues. Non-computability is not. **Reference:**

Lecture slides: Model virtues. Course literature: *Justified Method Choice*, modelling.

Q233. What is model robustness?

Mark all correct options.

a) Stability of results under variations in assumptions.
b) Perfect realism of the model.
c) A reason to have more confidence in a qualitative pattern.
d) Guarantee of correct prediction.

Correct answer(s):

a, c **Explanation:**

Robustness is stability under changes (a) and can support confidence in patterns (c). It is not realism (b false) and not a guarantee (d false). **Reference:**

Lecture slides: Robustness. Course literature: *Justified Method Choice*, modelling virtues.

Q234. What is model transparency?

Mark all correct options.

a) Understanding how assumptions produce results.
b) Having no assumptions.
c) Helping error detection and critique.
d) Being identical to accuracy.

Correct answer(s):

a, c **Explanation:**

Transparency concerns understandability and supports critique (a, c). Models always have assumptions (b false). Transparency differs from accuracy (d false). **Reference:**

Lecture slides: Transparency. Course literature: *Justified Method Choice*, modelling.

Q235. Which statements about correlation are correct?

Mark all correct options.

a) It measures association.
b) It is directional.
c) It can arise from common causes.
d) It is sufficient to infer causation.

Correct answer(s):

a, c **Explanation:**

Correlation measures association and can arise from common causes (a, c). It is symmetric, not directional (b false), and does not prove causation (d false). **Reference:**

Lecture slides: Correlation vs causation. Course literature: *Justified Method Choice*, causation chapter.

Q236. Which statements about causation are correct?

Mark all correct options.

a) It is typically asymmetric.
b) It can be investigated with experiments.
c) Correlation alone usually fixes causal direction.
d) Confounding can mislead causal inference.

Correct answer(s):

a, b, d **Explanation:**

Causation is directional (a) and experiments help identify it (b). Correlation alone usually does not determine direction (c false). Confounding can mislead (d). **Reference:**

Lecture slides: Causation; experiments. Course literature: *Justified Method Choice*, causation and validity.

Q237. Which statements about explanation structure are correct?

Mark all correct options.

a) The explanandum is what is explained.
b) The explanans is what does the explaining.
c) "Because" can mark the explanans following the explanandum.
d) Explanations never use background conditions.

Correct answer(s):

a, b, c **Explanation:**

Explanandum is the target phenomenon and explanans provides the explanation (a, b). "Because" often introduces explanans (c). Background conditions can be part of explanations (d false). **Reference:**

Lecture slides: Explanans/explanandum. Course literature: *Justified Method Choice*, explanation chapter.

Q238. Which claims about social-science anti-naturalism are correct?

Mark all correct options.

a) Interpretation is central for understanding human behavior.
b) Social science should mirror natural science methods in all respects.
c) Understanding can differ from causal explanation.
d) Social science cannot use evidence.

Correct answer(s):

a, c **Explanation:**

Anti-naturalists emphasize interpretation and distinguish understanding from causal explanation (a, c). They do not claim social science must mirror natural science (b false) nor that it cannot use evidence (d false). **Reference:**

Lecture slides: Anti-naturalism and interpretation. Course literature: *Justified Method Choice*, Ch. 9.

Q239. Which claims about belief-desire explanation are correct?

Mark all correct options.

a) It explains an action via attributed beliefs and desires.
b) It is identical to a physical law explanation.
c) It can be assessed for plausibility and fit with evidence.
d) It implies that ends justify means.

Correct answer(s):

a, c **Explanation:**

Belief-desire explanations cite mental states (a) and can be evaluated (c). They are not identical to physical-law explanations (b false) and do not entail a moral principle (d false). **Reference:**

Lecture slides: Belief-desire; intentional stance. Course literature: *Justified Method Choice*, Ch. 9.

Q240. Which are examples of tacit knowledge?

Mark all correct options.

a) Knowing how to weld well by practice.
b) Knowing a formal definition by heart.
c) Knowing how to tune an engine by sound and feel.
d) Knowing that Paris is in France.

Correct answer(s):

a, c **Explanation:**

Tacit knowledge is practical know-how hard to codify (a, c). Memorized facts and definitions are explicit (b, d false). **Reference:**

Lecture slides: Tacit knowledge. Course literature: *Justified Method Choice*, Ch. 8.

Q241. What is meant by "methodological naturalism" in social science debates?

Mark all correct options.

a) Social science can use similar methodological standards as natural science.
b) Interpretation is never relevant.
c) Explanation and prediction can be shared goals.
d) Only randomized experiments are scientific.

Correct answer(s):

a, c **Explanation:**

Naturalism supports shared methodological standards and goals (a, c). It need not deny interpretation (b false) and does not restrict science to randomized experiments (d false). **Reference:**

Lecture slides: Naturalism vs anti-naturalism. Course literature: *Justified Method Choice*, Ch. 9.

Q242. Which statements about methodological justification are correct?

Mark all correct options.

a) It can be relative to goals like prediction or explanation.
b) It is independent of practical constraints.
c) It can involve error management.
d) It never involves value judgments.

Correct answer(s):

a, c **Explanation:**

Justification is goal-relative and involves managing error (a, c). Practical constraints matter (b false). Value judgments can enter via goal choice and trade-offs (d false). **Reference:**

Lecture slides: Method choice and goals. Course literature: *Justified Method Choice*, methodology.

Q243. Which claims about uncertainty and decision-making are correct?

Mark all correct options.

a) Decision criteria can differ under risk vs deep uncertainty.
b) Assigning probabilities is always straightforward.
c) Utility functions can represent preferences under risk.
d) Expected value always captures what matters.

Correct answer(s):

a, c **Explanation:**

Risk differs from deep uncertainty in probability knowledge (a). Probabilities are not always straightforward (b false). Utility can represent preferences under risk (c). EV does not always capture what matters (d false). **Reference:**

Lecture slides: Risk and deep uncertainty. Course literature: *Justified Method Choice*, Ch. 12.

Q244. Which statements about error management are correct?

Mark all correct options.

a) It aims to reduce false conclusions and overconfidence.
b) It aims to eliminate all possibility of error.
c) It can involve setting acceptable error rates.
d) It is irrelevant to methodology.

Correct answer(s):

a, c **Explanation:**

Error management reduces error risks and sets acceptable error rates (a, c). Eliminating all error is unrealistic (b false). It is central to methodology (d false). **Reference:**

Lecture slides: Methodology as error management. Course literature: *Justified Method Choice*, methodology.

Q245. Which statements about "severe testing" are correct?

Mark all correct options.

a) A severe test would likely reveal flaws if the hypothesis were false.
b) Passing a severe test proves the hypothesis true.
c) Severe tests are connected to error detection.
d) Severity is unrelated to measurement quality.

Correct answer(s):

a, c **Explanation:**

Severity concerns the ability to detect errors if present (a, c). Passing does not prove truth (b false). Measurement quality affects severity (d false). **Reference:**

Lecture slides: Severe tests and error control. Course literature: *Justified Method Choice*, hypothesis testing.

Q246. Which statements about statistical significance are correct?

Mark all correct options.

a) It is not the same as practical importance.
b) It implies a large effect size.
c) It depends on sample size.
d) It can be misleading if assumptions are violated.

Correct answer(s):

a, c, d **Explanation:**

Significance differs from practical importance (a), depends on sample size (c), and can mislead when assumptions fail (d). It does not imply a large effect (b false). **Reference:**

Lecture slides: Statistical inference pitfalls. Course literature: *Justified Method Choice*, statistics chapter.

Q247. Which statements about reproducibility are correct?

Mark all correct options.

a) Clear reporting supports reproducibility.
b) Reproducibility guarantees truth.
c) Lack of reproducibility can indicate errors or instability.
d) Reproducibility is irrelevant in modelling.

Correct answer(s):

a, c **Explanation:**

Reporting supports reproducibility (a). Reproducibility does not guarantee truth (b false) but failures can signal problems (c). It matters for modelling too (d false). **Reference:**

Lecture slides: Reproducibility and transparency. Course literature: *Justified Method Choice*, methodology.

Q248. Which statements about replication are correct?

Mark all correct options.

a) Replication tests whether results recur.
b) Replication is pointless after publication.
c) Replication can reveal fragile findings.
d) Replication always uses identical procedures in identical contexts.

Correct answer(s):

a, c **Explanation:**

Replication checks recurrence and can reveal fragility (a, c). It is not pointless after publication (b false). Some replications vary contexts or procedures (d false). **Reference:**

Lecture slides: Replication. Course literature: *Justified Method Choice*, methodology.

Q249. Which statements about scientific models are correct?

Mark all correct options.

a) Models can be used to explore counterfactual scenarios.
b) Models can replace empirical evidence entirely.
c) Model adequacy is often purpose-relative.
d) Models can introduce errors if assumptions are poorly chosen.

Correct answer(s):

a, c, d **Explanation:**

Models explore "what-if" scenarios (a), are purpose-relative (c), and can mislead if assumptions are wrong (d). They do not replace empirical evidence entirely (b false). **Reference:**

Lecture slides: Models and purposes. Course literature: *Justified Method Choice*, modelling chapter.

Q250. Which statements about methodological criticism are correct?

Mark all correct options.

a) Methods can be criticized relative to goals and constraints.
b) A method cannot be criticized if it is common in a field.
c) Criticism can target error rates, validity, and relevance.
d) Criticism is always purely personal opinion.

Correct answer(s):

a, c **Explanation:**

Method criticism is goal-relative and can address error, validity, and relevance (a, c). Commonness does not immunize a method (b false). Criticism can be evidence-based, not merely personal (d false). **Reference:**

Lecture slides: Methodology and justification. Course literature: *Justified Method Choice*, methodology chapters.